Chapter 12

Theory of Games

"Good management is the art of making problems so interesting and their solutions so constructive that everyone wants to get to work and deal with them."

– Paul Hawken

PREVIEW

A game is a contest involving two or more competitors each of whom wants to win. A theory of games provides a series of mathematical models that may be useful in explaining interactive decision-making concepts, where two or more competitors are involved under conditions of conflict and competition.

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- understand how optimal strategies are formulated in conflict and competitive environment.
- understand the principles of two-person zero-sum games.
- apply various methods to select and execute various optimal strategies to win the game.
- use dominance rules to reduce the size of a game payoff matrix and compute value of the game with mixed strategies.
- apply minimax and maximin principle to compute the value of the game, when there is a saddle point.
- make distinction between pure and mixed strategies.
- use linear programming approach to compute the value of the game when dominance rules do not apply.

CHAPTER OUTLINE

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12.2 Two-Person Zero-Sum Games

12.3 Pure Strategies (Minimax and Maximin Principles): Games with Saddle Point

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12.1 INTRODUCTION

In general, the term 'game' refers to a situation of conflict and competition in which two or more competitors (or participants) are involved in the decision-making process in anticipation of certain outcomes over a period of time. The competitors are referred to as *players.* A player may be an *individual, individuals*, or an *organization.* A few examples of competitive and conflicting decision environment, that involve the interaction between two or more competitors are:

- Pricing of products, where sale of any product is determined not only by its price but also by the price set by competitors for a similar product
- The success of any TV channel programme largely depends on what the competitors presence in the same time slot and the programme they are telecasting.
- The success of a business strategy depends on the policy of internal revenue service regarding the expenses that may be disallowed,
- The success of an advertising/marketing campaign depends on various types of services offered to the customers.

Strategy in a course of action that a player adopts for every payoff (outcome).

For academic interest, *theory of games* provides a series of mathematical models that may be useful in explaining interactive decision-making concepts, where two or more competitors are involved under conditions of conflict and competition. However, such models provide an opportunity to a competitor to evaluate not only his personal decision alternatives (courses of action), but also the evaluation of the competitor's possible choices in order to win the game.

Game theory came into existence in 20th Century. However, in 1944 John Von Neumann and Oscar Morgenstern published a book named *Theory of Games and Economic Behavior*, in which they discussed how businesses of all types may use this technique to determine the best strategies given a competitive business environment. The author's approach was based on the principle of '*best out of the worst*'.

The models in the *theory of games* can be classified based on the following factors:

Number of players If a game involves only two players (competitors), then it is called a *two-person game*. However, if the number of players are more, the game is referred to as *n-person game*.

Sum of gains and losses If, in a game, the sum of the gains to one player is exactly equal to the sum of losses to another player, so that, the sum of the gains and losses equals zero, then the game is said to be a *zero-sum game*. Otherwise it is said to be *non-zero sum game*.

Strategy The strategy for a player is the list of all possible actions (moves, decision alternatives or courses of action) that are likely to be adopted by him for every payoff (outcome). It is assumed that the players are aware of the rules of the game governing their decision alternatives (or strategies). The outcome resulting from a particular strategy is also known to the players in advance and is expressed in terms of numerical values (e.g. money, per cent of market share or utility).

Pure strategy is the only course of action that is always chosen by a player.

The particular strategy that optimizes a player's gains or losses, without knowing the competitor's strategies, is called *optimal strategy.* The expected outcome, when players use their optimal strategy, is called *value of the game.*

Generally, the following two types of strategies are followed by players in a game:

(a) *Pure Strategy* A particular strategy that a player chooses to play again and again regardless of other player's strategy, is referred as *pure strategy*. The objective of the players is to maximize their gains or minimize their losses.

(b) *Mixed Strategy* A set of strategies that a player chooses on a particular move of the game with some fixed probability are called *mixed strategies*. Thus, there is a probabilistic situation and objective of the each player is to maximize expected gain or to minimize expected loss by making the choice among pure strategies with fixed probabilities.

Mathematically, if p_j (j = 1, 2, . . ., n) is the probability associated with a pure strategy j to be chosen by a player at any point in time during the game, then the set S of n non-negative real numbers

(probabilities) whose sum is unity associated with pure strategies of the player is written as: $S = \{ p_1, p_2, \ldots, p_n \}$ where $p_1 + p_2 + \ldots + p_n = 1$ and $p_j \geq 0$ of all j.

Remark If a particular $p_j = 1$ ($j = 1, 2, \ldots, n$) and all others are zero, the player is said to select pure strategy j. A flow chart of using game theory approach to solve a problem is shown in Fig. 12.1.

Mixed strategy are courses of action that are to be selected on a particular occasion with some fixed probability.

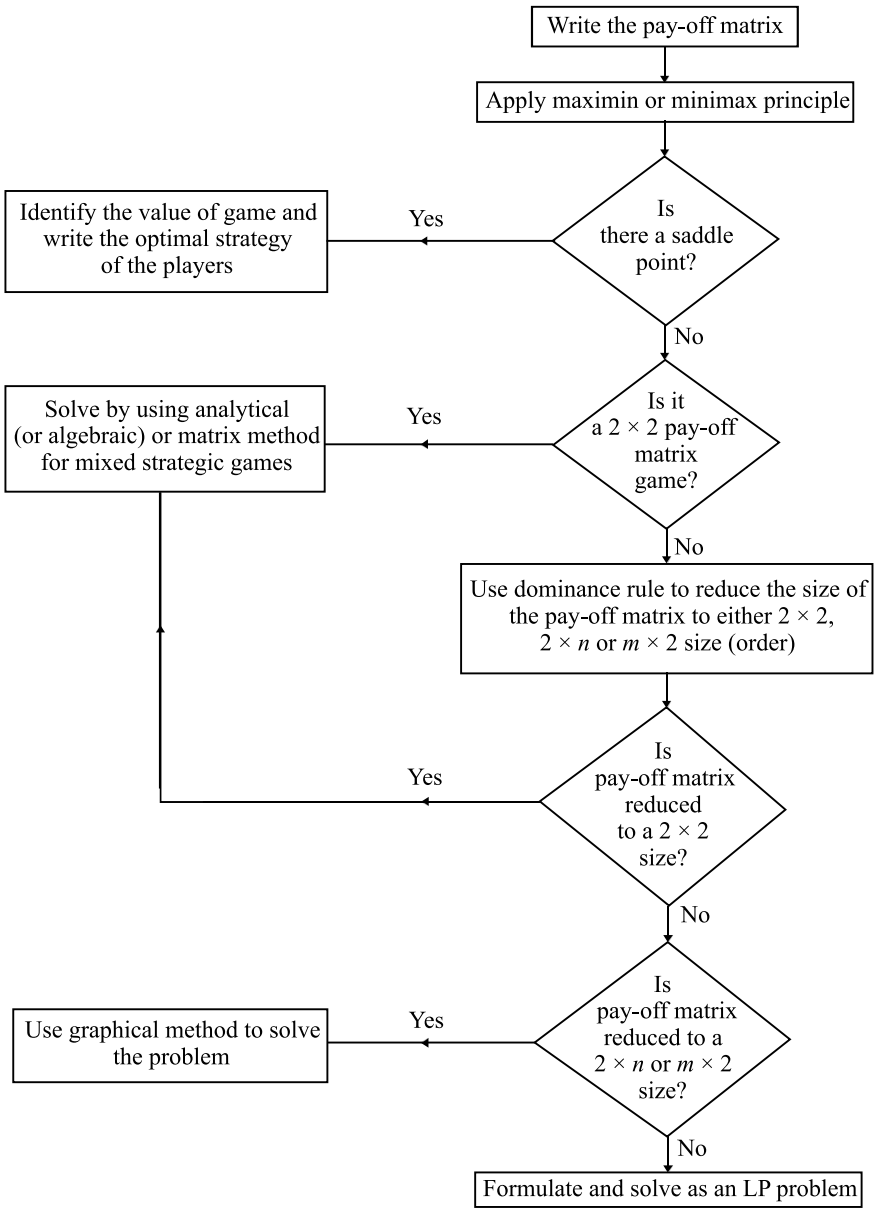


Fig. 12.1 Flow Chart of Game Theory Approach

12.2 TWO-PERSON ZERO-SUM GAMES

A game with only two players, say A and B, *is* called a *two-person zero-sum game*, only if one player's gain is equal to the loss of other player, so that total sum is zero.

Payoff matrix The payoffs (a quantitative measure of satisfaction that a player gets at the end of the play) in terms of gains or losses, when players select their particular strategies (courses of action), can be represented in the form of a matrix, called the payoff matrix. Since the game is zero-sum, the gain of one player is equal to the loss of other and vice versa. In other words, one player's payoff table would contain the same amounts in payoff table of other player, with the sign changed. Thus, it is sufficient to construct a payoff table only for one of the players.

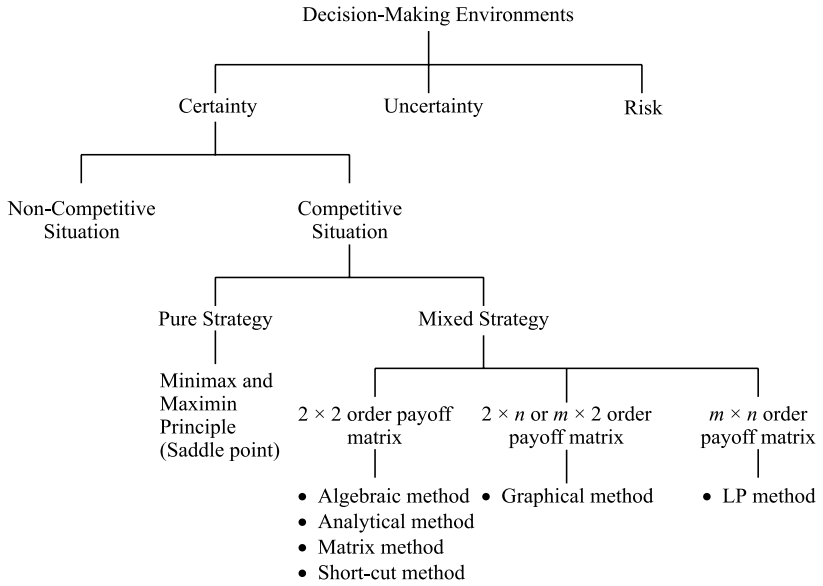
If player A has m strategies represented by the letters: $A_1, A_2, \ldots, A_m$ and player B has n strategies represented by the letters: $B_1, B_2, \ldots, B_n$. The numbers m and n need not be equal. The total number of possible outcomes is therefore $m \times n$. It is assumed that each player not only knows his own list of possible strategies but also of his competitor. For convenience, it is assumed that player A is always a gainer whereas player B a loser. Let a_{ij} be the payoff that player A gains from player B if player A chooses strategy i and player B chooses strategy j. Then the payoff matrix is shown in the Table 12.1.

Player A's Strategies	Player B's Strategies			
	B_1	B_2	...	B_n
A_1	a_{11}	a_{12}	...	a_{1n}
A_2	a_{21}	a_{22}	...	a_{2n}
⋮	⋮	⋮		⋮
A_m	a_{m1}	a_{m2}	...	a_{mn}

Table 12.1
Payoff Matrix

Since player A is assumed to be the gainer, therefore he wishes to gain as large a payoff a_{ij} as possible, player B on the other hand would do his best to reach as small a value of a_{ij} as possible. Of course, the gain to player B and loss to A must be $-a_{ij}$.

Various methods discussed in this chapter to find value of the game under decision-making environment of certainty are as follows:



Assumptions of the game

1. Each player has available to him a finite number of possible strategies (courses of action). The list may not be the same for each player.
2. Players act rationally and intelligently.
3. List of strategies of each player and the amount of gain or loss on an individual's choice of strategy is known to each player in advance.
4. One player attempts to maximize gains and the other attempts to minimize losses.
5. Both players make their decisions individually, prior to the play, without direct communication between them.
6. Both players select and announce their strategies simultaneously so that neither player has an advantage resulting from direct knowledge of the other player's decision.
7. The payoff is fixed and determined in advance.

12.3 PURE STRATEGIES (MINIMAX AND MAXIMIN PRINCIPLES): GAMES WITH SADDLE POINT

Maximin principle maximize the player's minimum gains.

The selection of an optimal strategy by each player, without the knowledge of the competitor's strategy, is the basic problem of playing games. Since the payoffs for either player provides all the essential information, therefore, only one player's payoff table is required to evaluate the decisions. By convention, the payoff table for the player whose strategies are represented by rows (say player A) is constructed. The objective of the study is to know how these players must select their respective strategies so that they are able to optimize their payoff. Such a decision-making criterion is referred to as the *minimax-maximin principle.* Such principle in pure strategies game always leads to the best possible selection of a strategy for both players.

Maximin principle For player A the minimum value in each row represents the least gain (payoff) to him, if he chooses his particular strategy. These are written in the matrix by row minima. He will then select the strategy that gives the largest gain among the row minimum values. This choice of player A is called the *maximin principle,* and the corresponding gain is called the *maximin value of the game.*

Minimax principle For player B, who is assumed to be the looser, the maximum value in each column represents the maximum loss to him, if he chooses his particular strategy. These are written in the payoff matrix by column maxima. He will then select the strategy that gives the minimum loss among the column maximum values. This choice of player B is called the *minimax principle,* and the corresponding loss is the *minimax value of the game.*

Value of the game is the expected gain or loss in a game when a game is played a large number of times.

Optimal strategy A course of action that puts any player in the most preferred position, irrespective of the course of action his competitor(s) adopt, is called as optimal strategy. In other words, if the maximin value equals the minimax value, then the game is said to have a *saddle* (*equilibrium*) *point* and the corresponding strategies are called *optimal strategies.*

Value of the game This is the expected payoff at the end of the game, when each player uses his optimal strategy, i.e. the amount of payoff, V, at an equilibrium point. A game may have more than one saddle points. A game with no saddle point is solved by choosing strategies with fixed probabilities.

Remarks 1. The value of the game, in general, satisfies the equation: maximin value $\leq V \leq$ minimax value.
2. A game is said to be *a fair* game if the lower (maximin) and upper (minimax) values of the game are equal and both equals zero.
3. A game is said to be *strictly determinable* if the lower (maximin) and upper (minimax) values of the game are equal and both equal the value of the game.

12.3.1 Rules to Determine Saddle Point

The reader is advised to follow the following three steps, in this order, to determine the saddle point in the payoff matrix.

Saddle point is the payoff value that represents both minimax and maximin value of the game.

1. Select the minimum (lowest) element in each row of the payoff matrix and write them under 'row minima' heading. Then, select the largest element among these elements and enclose it in a rectangle, $\boxed{}$.
2. Select the maximum (largest) element in each column of the payoff matrix and write them under 'column maxima' heading. Then select the lowest element among these elements and enclose it in a circle, $\bigcirc$.
3. Find out the element(s) that is same in the circle as the well as rectangle and mark the position of such element(s) in the matrix. This element represents the value of the game and is called the saddle (or equilibrium) point.

Example 12.1 For the game with payoff matrix:

	Player B		
Player A	B_1	B_2	B_3
A_1	– 1	2	– 2
A_2	6	4	– 6

determine the optimal strategies for players A and B. Also determine the value of game. Is this game (i) fair? (ii) strictly determinable?

Solution In this example, gains to player A or losses to player B are represented by the positive quantities, whereas, losses to A and gains to B are represented by negative quantities. It is assumed that A wants to

maximize his minimum gains from B. Since the payoffs given in the matrix are what A receives, therefore, he is concerned with the quantities that represent the row minimums. Now A can do no worse than receive one of these values. The best of these values occurs when he chooses strategy A_1. This choice provides a payoff of – 2 to A when B chooses strategy B_3. This refers to A's choice of A_1 as his maximum payoff strategy because this row contains the maximum of A's minimum possible payoffs from his competitor B.

	Player B			
Player A	B_1	B_2	B_3	Row minimum
A_1	–1	2	(– 2)	[– 2] ← *Maximin*
A_2	6	4	– 6	– 6
Column maximum	6	4	(– 2) ← *Minimax*	

Table 12.2

Similarly, it is assumed that B wants to minimize his losses and wishes that his losses to A be as small as possible. The column maximums also represent the greatest payments B might have to make to A. The smallest of these losses is – 2, which occurs when A chooses his course of action, A_1 and B chooses his course of action, B_3. This choice of B_3 by B is his minimax loss strategy because the amount of this column is the minimum of the maximum possible losses.

In Table 12.2, the quantity – 2 in row A_1 and column B_3 is enclosed both in the box and the circle. That is, it is both the minimum of the column maxima and the maximum of the row minima. This value is referred to as *saddle point*.

The payoff amount in the saddle-point position is also called *value of the game*. For this game, value of the game is, $V = -2$, for player A. The value of game is always expressed from the point of view of the player whose strategies are listed in the rows.

The game is strictly determinable. Also since the value of the game is not zero, the game is not fair.

Example 12.2 A company management and the labour union are negotiating a new three year settlement. Each of these has 4 strategies:

I : Hard and aggressive bargaining II : Reasoning and logical approach

III : Legalistic strategy IV : Conciliatory approach

The costs to the company are given for every pair of strategy choice.

	Company Strategies			
Union Strategies	*I*	*II*	*III*	*IV*
I	20	15	12	35
II	25	14	8	10
III	40	2	10	5
IV	– 5	4	11	0

What strategy will the two sides adopt? Also determine the value of the game.

Solution Applying the rule of finding out the saddle point, we obtain the saddle point that is enclosed both in a circle and a rectangle, as shown in Table 12.3.

	Company Strategies				
Union Strategies	*I*	*II*	*III*	*IV*	Row minimum
I	20	15	(12)	35	[12] ← *Maximin*
II	25	14	8	10	8
III	40	2	10	5	2
IV	– 5	4	11	0	– 5
Column maximum	40	15	(12)	35	
			↑ *Minimax*		

Table 12.3

As shown in Table 12.3, since Maximin = Minimax = Value of game = 12, therefore the company will always adopt strategy III – Legalistic strategy and union will always adopt strategy I – Hard and aggressive bargaining.

Example 12.3 Find the range of values of p and q that will render the entry (2, 2) a saddle point for the game:

	Player B		
Player A	B_1	B_2	B_3
A_1	2	4	5
A_2	10	7	q
A_3	4	p	6

Solution First ignore the values of p and q in the payoff matrix, and then determine the maximin and minimax values in the usual manner, as shown in Table 12.4.

Table 12.4

	Player B			
Player A	B_1	B_2	B_3	Row minimum
A_1	2	4	5	2
A_2	10	[7]	q	[7] ← *Maximin*
A_3	4	p	(6)	4
Column maximum	10	7	(6)	
			↑ *Minimax*	

As shown in Table 12.4, since there exists no unique saddle point, therefore, the saddle point will exist at the position (2, 2) only when $p \leq 7$ and $q > 7$.

Example 12.4 For what value of λ, the game with following pay-off matrix is strictly determinable?

	Player B		
Player A	B_1	B_2	B_3
A_1	λ	6	2
A_2	-1	λ	-7
A_3	-2	4	λ

Solution First, ignoring the value of λ, determine the maximin and minimax values of the pay-off matrix, as shown below:

	Player B			
	B_1	B_2	B_3	Row minimum
A_1	λ	6	[2]	(2) ← *Maximin*
A_2	(–1)	λ	–7	–7
A_3	–2	4	λ	–2
Column maximum	(–1)	6	2	
	↑ *Minimax*			

Since saddle point in the above table is not unique, the value of the game lies between –1 and 2, i.e. $-1 \leq V \leq 2$. For strictly determinable game, we must have $-1 \leq \lambda \leq 2$.

CONCEPTUAL QUESTIONS A

1. Define: (i) competitive game, (ii) payoff matrix, (iii) pure and mixed strategies, (iv) saddle point, (v) optimal strategies, and (vi) rectangular (or two-person zero-sum) game.
2. Explain: Minimax and Maximin principle used in the theory of games.
3. What is a game in game theory? What are the properties of a game? Explain the 'best strategy' on the basis of minimax criterion of optimality. [*Delhi Univ., MBA, 2006*]
4. Explain the two-person zero-sum game, giving a suitable example.
5. Define 'saddle point'. Is it necessary that a game should always possess a saddle point? [*Agra MCA, 2000*]
6. Let $A = [a_{ij}]$ be the payoff matrix for a two-person zero-sum game. If V denotes the minimax value of the game, then show that $V \geq \bar{V}$. [*Meerut MSc (Stat.), 2002*]
7. Define: (i) Competitive game; (ii) Pure strategies; (iii) Mixed strategies (iv)Two-person zero-sum (or rectangular) game, (v) Payoff matrix.
8. State the major limitations of the game theory. [*Delhi Univ., MCom, 2000*]

9. Explain the difference between pure strategy and mixed strategy.
10. Which competitive situation is called a game? What is the maximin criterion of optimality?
11. What are the assumptions made in the theory of games?
12. Describe the maximin principle of game theory. What do you understand by pure strategies, and 'saddle point'.

[*Punjab, Univ. MBA 2001*]

13. Game theory provides a systematic quantitative approach for analysing competitive situations in which the competitors make use of logical processes and techniques in order to determine an optimal strategy for winning: Comment.

[*Delhi Univ., MBA, 2003*]

SELF PRACTICE PROBLEMS A

1. Consider the game with the following payoff table:

	Player *B*	
Player *A*	B_1	B_2
A_1	2	6
A_2	– 2	λ

(a) Show that the game is strictly determinable, whatever λ may be.
(b) Determine the value of the game.

2. Determine which of the following two-person zero-sum games are strictly determinable and fair. Give the optimum strategies for each player in the case of it being strictly determinable.

(a)

Player *A*	Player *B*	
	B_1	B_2
A_1	1	2
A_2	4	– 3

(b)

Player *A*	Player *B*	
	B_1	B_2
A_1	– 5	2
A_2	– 7	– 4

3. Solve the following games by using maximin (minimax) principle, whose payoff matrix are given below: Include in your answer: (i) strategy selection for each player, (ii) the value of the game to each player. Does the game have a saddle point?

(a)

	Player *B*			
Player *A*	B_1	B_2	B_3	B_4
A_1	1	7	3	4
A_2	5	6	4	5
A_3	7	2	0	3

(b)

	Player *B*				
Player *A*	B_1	B_2	B_3	B_4	B_5
A_1	– 2	0	0	5	3
A_2	3	2	1	2	2
A_3	– 4	– 3	0	– 2	6
A_4	5	3	– 4	2	6

(c)

	Player *B*			
Player *A*	B_1	B_2	B_3	B_4
A_1	3	– 5	0	6
A_2	– 4	– 2	1	2
A_3	5	4	2	3

4. (a)

	Player *B*		
Player *A*	B_1	B_2	B_3
A_1	– 2	15	– 2
A_2	– 5	– 6	– 4
A_3	– 5	20	– 8

(b)

	Player *B*			
Player *A*	B_1	B_2	B_3	B_4
A_1	– 5	3	1	10
A_2	5	5	4	6
A_3	4	– 2	0	– 5

5.

	Firm *B*				
Firm *A*	B_1	B_2	B_3	B_4	B_5
A_1	3	– 1	4	6	7
A_2	– 1	8	2	4	12
A_3	16	8	6	14	12
A_4	1	11	– 4	2	1

6. Two competitive manufacturers are producing a new toy under licence from a patent holder. In order to meet the demand they have the option of running the plant for 8, 16 or 24 hours a day. As the length of production increases so does the cost. One of the manufacturers, say A, has set up the matrix given below. He uses the matrix to estimate the percentage of the market that he could capture and maintain the different production schedules:

	Manufacturer *B*		
Manufacturer *A*	C_1 : 8 hrs	C_2 : 16 hrs	C_3 : 24 hrs
S_1 : 8 hrs	60%	56%	34%
S_2 : 16 hrs	63%	60%	55%
S_3 : 24 hrs	83%	72%	60%

(i) At which level should each produce?
(ii) What percentage of the market will *B* have?

7. Two computer manufacturers *A* and *B* are attempting to sell computer systems to two banks 1 and 2. Company *A* has 4 salesmen, company *B* only has 3 salesmen available. The computer companies must decide upon how many salesmen to assign to sell computer to each bank. Thus company *A* can assign 4 salesmen to bank 1 and none to bank 2 or three to bank 1 and one to bank 2, etc.

Each bank will buy one computer system. The probability that a bank will buy from a particular computer company is directly related to the number of salesmen calling from that company, relative to the total salesmen calling. Thus, if company *A* assigns three salesmen to bank 1 and company *B* assigns two salesmen, the odds would be three out of five that bank 1 would purchase

company *A*'s computer system. (If none calls from either company the odds are one-half for buying either computer.)

Let the payoff be the expected number of computer systems that company *A* sells. (2 minus this payoff is the expected number company *B* sells).

What strategy would company *A* use in allocating its salesmen? What strategy should company *B* use? What is the value of the game to company *A*? What is the meaning of the value of the game in this problem?

8. Assume that two firms are competing for the market share for a particular product. Each firm is considering what promotional strategy to employ for the coming period. Assume that the following payoff matrix describes the increase in market share of Firm *A* and the decrease in market share for Firm *B*. Determine the optimal strategies for each firm.

	Firm *B*		
Firm *A*	*No Promotion Promotion*	*Moderate Promotion*	*Much*
No Promotion	5	0	–10
Moderate Promotion	10	6	2
Much Promotion	20	15	10

(i) Which firm would be the winner, in terms of market share?
(ii) Would the solution strategies necessarily maximize the profits for either of the firms?
(iii) What might the two firms do to maximize their profits?

9. The XYZ and ABC are both a chain of chinese foods restaurants located in Delhi. Both chains are financially strong enough to expand. The only viable manner in which this may be accomplished is for each chain to open stores in other cities. The following payoff (in Rs) tables provide the anticipated average profit levels over the next five years for the various courses of action.

XYZ's Profits

	ABC doesn't expand	*ABC expands*
XYZ doesn't expand	5,00,000	– 2,00,000
XYZ expands	8,00,000	2,00,000

ABC's Profits

	ABC doesn't expand	*ABC expands*
XYZ doesn't expand	4,00,000	5,00,000
XYZ expands	– 4,00,000	60,000

(a) Is this situation a zero-sum game? Explain.
(b) Find the equilibrium courses of action, if any.
(c) If the two firms can cooperate, what course of action would each prefer?

HINTS AND ANSWERS

1. (a) Ignoring the value of λ, the maximin value = 2 = minimax. Thus the game is strictly determinable.
   (b) Value of game is 2 with optimal strategy for player *A* and *B* is A_1 and B_1, respectively.
2. (i) Not fair, value of game = 1
   (ii) Not fair, value of game = – 5
3. (a) Optimal strategy, $A : A_2$; $B : B_3$, Value of game = 4
   (b) Optimal strategy, $A : A_2$; $B : B_3$, Value of game = 1
   (c) Optimal strategy, $A : A_3$; $B : B_3$, Value of game = 2
4. (a) Two points at (A_1, B_1), (A_1, B_3), Value of game = – 2
   (b) Optimal strategies: (A_2, B_3), Value of game = 4
5. Optimal strategy: (A_3, B_3), Value of game = 6
6. (i) Optimal strategy: (S_3, C_3), i.e. both should produce at the level of 24 hours per day.
   (ii) At the optimal level of 24 hours per day, *B* will have 60 per cent of the market share.
7. Payoff matrix for company *A*

	Company *B*			
Company *A*	B_1	B_2	B_3	B_4
A_1	1/2	0	0	0
A_2	1	1/2	1/3	1/4
A_3	1	2/3	2/4	2/5
A_4	1	3/5	3/5	3/6
A_5	1	4/5	4/6	4/7

Optimal strategy for *A* is A_5 and for *B* is B_4. Value of game = 4/7, i.e. probability of success for *A* is 4/7 or 51 per cent supply.
8. Optimal strategy for both *A* and *B* is 'Much promotion'; Value of game = 10.

12.4 MIXED STRATEGIES: GAME WITHOUT SADDLE POINT

In certain cases, no saddle point exists, i.e. maximin value ≠ minimax value. In all such cases, players must choose the mixture of strategies to find the value of game and an optimal strategy.

The value of game obtained by the use of mixed strategies represents the least payoff, which player *A* can expect to win and the least which player *B* can expect to lose. The expected payoff to a player in a game with payoff matrix $[a_{ij}]$ of order $m \times n$ is defined as:

$$E(p, q) = \sum_{i=1}^{m} \sum_{j=1}^{n} p_i \, a_{ij} \, q_j = \mathbf{P}^T \mathbf{A} \, \mathbf{Q} \text{ (in matrix notation)}$$

where $\mathbf{P} = (p_1, p_2, \ldots, p_m)$ and $\mathbf{Q} = (q_1, q_2, \ldots, q_n)$ denote probabilities (or relative frequency with which a strategy is chosen from the list of strategies) associated with m strategies of player A and n strategies of player, B respectively, where $p_1 + p_2 + \ldots + p_m = 1$ and $q_1 + q_2 + \ldots + q_n = 1$.

A mixed strategy game can be solved by using following methods:

- *Algebraic method*
- *Matrix method*
- *Linear programming method.*
- *Analytical or calculus method*
- *Graphical method, and*

Remark For solving a 2×2 game, without a saddle point, the following formula is also used. If payoff matrix for player A is given by:

$$\begin{array}{c} \\ \text{Player } A \end{array} \begin{array}{c} \text{Player } B \\ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \end{array}$$

then the following formulae are used to find the value of game and optimal strategies:

$$p_1 = \frac{a_{22} - a_{21}}{a_{11} + a_{22} - (a_{12} + a_{21})}; \quad q_1 = \frac{a_{22} - a_{12}}{a_{11} + a_{22} - (a_{12} + a_{21})}$$

where $p_2 = 1 - p_1; \quad q_2 = 1 - q_1$

and $V = \dfrac{a_{11}a_{22} - a_{21}a_{12}}{a_{11} + a_{22} - (a_{12} + a_{21})}$

12.5 THE RULES (PRINCIPLES) OF DOMINANCE

The rules of dominance are used to reduce the size of the payoff matrix. These rules help in deleting certain rows and/or columns of the payoff matrix that are inferior (less attractive) to at least one of the remaining rows and/or columns (strategies), in terms of payoffs to both the players. Rows and/or columns once deleted can never be used for determining the optimum strategy for both the players.

Dominance rules is the procedure to reduce the size of the payoff matrix according to the tendency of players.

The rules of dominance are especially used for the evaluation of two-person zero-sum games without a saddle (equilibrium) point. Certain dominance principles are stated as follows:

1. For player B, who is assumed to be the loser, if each element in a column, say C_r is greater than or equal to the corresponding element in another column, say C_s in the payoff matrix, then the column C_r is said to be dominated by column C_s and therefore, column C_r can be deleted from the payoff matrix. In other words, *player B will never use the strategy that corresponds to column C_r because he will loose more by choosing such strategy*.
2. For player A, who is assumed to be the gainer, if each element in a row, say R_r, is less than or equal to the corresponding element in another row, say R_s, in the payoff matrix, then the row R_r is said to be dominated by row R_s and therefore, row R_r can be deleted from the payoff matrix. In other words, *player A will never use the strategy corresponding to row R_r, because he will gain less by choosing such a strategy*.
3. A strategy say, k can also be dominated if it is inferior (less attractive) to an average of two or more other pure strategies. In this case, if the domination is strict, then strategy k can be deleted. If strategy k dominates the convex linear combination of some other pure strategies, then one of the pure strategies involved in the combination may be deleted. The domination would be decided as per rules 1 and 2 above.

Remark Rules (principles) of dominance discussed are used when the payoff matrix is a profit matrix for the player A and a loss matrix for player B. Otherwise the principle gets reversed.

Example 12.5 Players A and B play a game in which each has three coins, a 5*p*, 10*p* and a 20*p*. Each selects a coin without the knowledge of the other's choice. If the sum of the coins is an odd amount, then A wins B's coin. But, if the sum is even, then B wins A's coin. Find the best strategy for each player and the values of the game. [*Agra Univ., MCA, 2000*]

Solution The payoff matrix for player A is

	Player B		
Player A	$5p : B_1$	$10p : B_2$	$20p : B_3$
$5p : A_1$	– 5	10	20
$10p : A_2$	5	– 10	– 10
$20p : A_3$	5	– 20	– 20

It is clear that this game has no saddle point. Therefore, further we must try to reduce the size of the given payoff matrix as further as possible. Note that every element of column B_3 (strategy B_3 for player B) is more than or equal to every corresponding element of row B_2 (strategy B_2 for player B). Evidently, the choice of strategy B_3, by the player B, will always result in more losses as compared to that of selecting the strategy B_2. Thus, strategy B_3 is inferior to B_2. Hence, delete the B_3 strategy from the payoff matrix. The reduced payoff matrix is shown below:

	Player B		
Player A	B_1	B_2	B_3
A_1	–5	10	20
A_2	5	– 10	– 10
A_3	5	– 20	– 20

After column B_3 is deleted, it may be noted that strategy A_2 of player A is dominated by his A_3 strategy, since the profit due to strategy A_2 is greater than or equal to the profit due to strategy A_3, regardless of which strategy player B selects. Hence, strategy A_3 (row 3) can be deleted from further consideration. Thus, the reduced payoff matrix becomes:

	Player B		
Player A	B_1	B_2	Row minimum
A_1	[– 5]	10	[– 5] ← *Maximin*
A_2	(5)	–10	–10
Column maximum	(5)	10	
	↑ *Minimax*		

As shown in the reduced 2×2 matrix, the maximin value is not equal to the minimax value. Hence, there is no saddle point and one cannot determine the point of equilibrium. For this type of game situation, it is possible to obtain a solution by applying the concept of mixed strategies.

The solution to this game can now be obtained by applying any of the methods used for mixed-strategy games that were discussed later.

12.6 SOLUTION METHODS FOR GAMES WITHOUT SADDLE POINT

12.6.1 Algebraic Method

This method is used to determine the probability of using different strategies by players A and B. This method becomes quite lengthy when a number of strategies for both the players are more than two.

Consider a game where the payoff matrix is: $[a_{ij}]_{m \times n}$. Let $(p_1, p_2, \ldots, p_m)$ and $(q_1, q_2, \ldots, q_n)$ be the probabilities with which players A and B select their strategies $(A_1, A_2, \ldots, A_m)$ and $(B_1, B_2, \ldots, B_n)$, respectively. If V is the value of game, then the expected gain to player A, when player B selects strategies $B_1, B_2, \ldots, B_n$, one by one, is given by left-hand side of the following simultaneous equations, respectively. Since player A is the gainer player and expects at least V, therefore, we must have

	Player B				
Player A	B_1	B_2	...	B_n	*Probability*
A_1	a_{11}	a_{12}	...	a_{1n}	p_1
A_2	a_{21}	a_{22}	...	a_{2n}	p_2
⋮	⋮				⋮
A_m	a_{m1}	a_{m2}	...	a_{mn}	p_m
Probability	q_1	q_2	...	q_n	

$$\begin{aligned} a_{11}\,p_1 + a_{21}\,p_2 + \ldots + a_{m1}\,p_m &\geq V \\ a_{12}\,p_1 + a_{22}\,p_2 + \ldots + a_{m2}\,p_m &\geq V \\ \vdots \qquad\quad \vdots \qquad\quad \vdots & \\ a_{1n}\,p_1 + a_{2n}\,p_2 + \ldots + a_{mn}\,p_m &\geq V \end{aligned} \tag{1}$$

where $p_1 + p_2 + \ldots + p_m = 1$ and $p_i \geq 0$ for all i

Algebraic method is used to determine the probability of using different strategies by players *A* and *B*.

Similarly, the expected loss to player B, when player A selects strategies A_1, A_2, ..., A_m, one by one, can also be determined. Since player B is the loser player, therefore, he must have:

$$\begin{aligned} a_{11}\,q_1 + a_{12}\,q_2 + \ldots + a_{1n}\,q_n &\leq V \\ a_{21}\,q_1 + a_{22}\,q_2 + \ldots + a_{2n}\,q_n &\leq V \\ \vdots \qquad\quad \vdots \qquad\quad \vdots & \\ a_{m1}\,q_1 + a_{m2}\,q_2 + \ldots + a_{mn}\,q_n &\leq V \end{aligned} \tag{2}$$

where $q_1 + q_2 + \ldots + q_n = 1$ and $q_j \geq 0$ for all j

To get the values of p_i's and q_j's, the above inequalities are considered as equations and are then solved for given unknowns. However, if the system of equations, so obtained, is inconsistent, then at least one of the inequalities must hold as a strict inequality. The solution can now be obtained only by applying the trial and error method.

Example 12.6 A company is currently involved in negotiations with its union on the upcoming wage contract. Positive signs in table represent wage increase while negative sign represents wage reduction. What are the optimal strategies for the company as well as the union? What is the game value?

Conditional costs to the company (Rs. in lakhs)

		Union Strategies			
		U_1	U_2	U_3	U_4
Company Strategies	C_1	0.25	0.27	0.35	−0.02
	C_2	0.20	0.06	0.08	0.08
	C_3	0.14	0.12	0.05	0.03
	C_4	0.30	0.14	0.19	0.00

[*Delhi Univ.*, *MBA*, *2002*]

Solution Suppose, Company is the gainer player and Union is the looser player. Transposing payoff matrix because company's interest is to minimize the wage increase while union's interest is to get the maximum wage increase.

		Company Strategies			
		C_1	C_2	C_3	C_4
Union Strategies	U_1	0.25	0.20	0.14	0.30
	U_2	0.27	0.16	0.12	0.14
	U_3	0.35	0.08	0.15	0.19
	U_4	−0.02	0.08	0.13	0.00

In this payoff matrix strategy U_4 is dominated by strategy U_1 as well as U_3. After deleting this strategy, we get

		Company Strategies			
		C_1	C_2	C_3	C_4
Union Strategies	U_1	0.25	0.20	0.14	0.30
	U_2	0.27	0.16	0.12	0.14
	U_3	0.35	0.08	0.15	0.19

Company's point of view, strategy C_1 is dominated by C_2 as well as C_3, while C_4 is dominated C_3. Deleting strategies C_1 and C_4 we get

		Company Strategies	
		C_2	C_3
Union Strategies	U_1	0.20	0.14
	U_2	0.16	0.12
	U_3	0.08	0.15

Again strategy U_2 is dominated by U_1 and is, therefore, deleted to give

		Company Strategies		
		C_2	C_3	*Probability*
Union Strategies	U_1	0.20	0.14	0.07/0.13 = 0.538
	U_3	0.08	0.15	0.06/0.13 = 0.461
Probability		0.01/0.13 = 0.076	0.12/0.13 = 0.923	

Optimal strategy for the company : (0, 0.076, 0.923, 0)
Optimal strategy for the union : (0.538, 0, 0.461, 0)
Value of the game, V : $0.538 \times 0.20 + 0.461 \times 0.08 =$ Rs. 14360

Example 12.7 In a game of matching coins with two players, suppose A wins one unit of value when there are two heads, wins nothing when there are two tails and losses 1/2 unit of value when there is one head and one tail. Determine the payoff matrix, the best strategies for each player and the value of the game to A.

Solution The payoff matrix for the given matching coin games is given by:

	Player B	
Player A	B_1	B_2
A_1	1	– 1/2
A_2	– 1/2	0

As the payoff matrix does not have a saddle point, the game will be solved by the algebraic method.

For Player A Let p_1 and p_2 be probabilities of selecting strategy A_1 and A_2, respectively. Then the expected gain to player A, when player B uses its B_1 and B_2 strategies, respectively, is given by:

$$p_1 - (1/2)\, p_2 \geq V; \quad B \text{ selects } B_1 \text{ strategy} \tag{1}$$

$$-(1/2)\, p_2 + 0.\, p_2 \geq V; \quad B \text{ selects } B_2 \text{ strategy} \tag{2}$$

where
$$p_1 + p_2 = 1 \tag{3}$$

For obtaining value of p_1 and p_2, we consider inequalities (1) and (2) as equations and then with the help of Eq. (3), we get $p_1 = -2V$ and $p_2 = -6V$.

Substituting these values of p_1 and p_2 in Eq. (3) we get $V = -1/8$. Thus, $p_1 = 0.25$ and $p_2 = 0.75$.

For Player B Let q_1 and q_2 be the probabilities of selecting strategies B_1 and B_2, respectively. Then the expected loss to player B, when player A uses its A_1 and A_2 strategies, respectively is given by:

$$q_1 - (1/2).q_2 \leq V;\ A \text{ selects } A_1 \text{ strategy} \tag{4}$$

$$-(1/2).q_1 + 0.q_2 \leq V;\ A \text{ selects } A_2 \text{ strategy} \tag{5}$$

and
$$q_1 + q_2 = 1 \tag{6}$$

Consider inequalities (4) and (5) as equations and then, with the help of Eq. (6), we get $q_1 = 2V$ and $q_2 = -6V$.

Substituting values of q_1 and q_2 in Eq. (6), we get $V = -1/8$. Thus, $q_1 = 0.25$ and $q_2 = 0.75$. Hence, the probability of selecting strategies optimally for players A and B are (0.25, 0.75) and (0.25, 0.75), respectively, and the value of the game is $V = -1/8$.

Alternative methods

(a) Using the following results, the values of p_1, p_2, q_1, q_2 and value of game, V can be calculated as follows:

For Player A

$$p_1 = \frac{a_{22} - a_{21}}{a_{11} + a_{22} - (a_{12} + a_{21})} = \frac{0 - (-1/2)}{1 + 0 - (-1/2 - 1/2)} = \frac{1}{4}, \text{ and}$$

$$p_2 = 1 - p_1 = \left(1 - \frac{1}{4}\right) = \frac{3}{4}$$

For Player B

$$q_1 = \frac{a_{22} - a_{12}}{a_{11} + a_{22} - (a_{12} + a_{21})} = \frac{0 - (-1/2)}{1 + 0 - (-1/2 - 1/2)} = \frac{1}{4}, \text{ and}$$

$$q_2 = 1 - q_1 = \left(1 - \frac{1}{4}\right) = \frac{3}{4}$$

Expected value of game, $V = \dfrac{a_{11}.a_{22} - a_{21}.a_{12}}{a_{11} + a_{22} - (a_{12} + a_{21})} = \dfrac{1 \times 0 - (-1/2) \times (-1/2)}{1 + 0 - (-1/2 - 1/2)} = -\dfrac{1}{8}$

(b) Expected payoff function

$E(p, q)$ = {(Expected gain to player A when player B uses strategy B_1) × Probability of player B using strategy B_1} + {(Expected gain to player A when player B uses strategy B_2) × Probability of player B using strategy B_2}

$= p_1 q_1 - (1/2) p_2 q_1 - (1/2) p_1 q_2 + (0) p_2 q_2$

$= p_1 q_1 - (1/2)(1 - p_1) q_1 - (1/2) p_1 (1 - q_1)$

$= p_1 q_1 - (1/2) q_1 + (1/2). p_1 q_1 - (1/2) p_1 + (1/2) p_1 q_1$

Differentiating partially $E(p, q)$ with respect to p_1 and q_1 and by equating with zero, we get

$$\frac{\partial E}{\partial p_1} = q_1 - (1/2) q_1 - (1/2) + (1/2) q_1 = 0, \text{ i.e. } q_1 = (1/4) \text{ and } q_2 = 1 - q_1 = (3/4)$$

$$\frac{\partial E}{\partial q_1} = p_1 - (1/2) + (1/2) p_1 + (1/2) p_1 = 0, \text{ i.e. } p_1 = (1/4) \text{ and } p_2 = 1 - p_1 = (3/4)$$

Substituting value of p_1 and q_1 in $E(p, q)$, we get:

Expected value of game $V = (1/4)(1/4) - 1/2\ (3/4)(1/4) - 1/2\ (1/4)(3/4)$

$= (1/16) - (3/32) - (3/32) = -1/8$

Example 12.8 Solve the game whose payoff matrix is given below:

	Player B			
Player A	B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

[*Punjab Univ., M Com 2003, 2006; Karn Univ, BE* (*Mech*)*, 2004*]

Solution It is clear that this game has no saddle point. Therefore, we try to reduce the size of the given payoff matrix by using dominance principles.

From player A's point of view, the first row is dominated by the third row, yielding the reduced 3 × 4 payoff matrix. In the reduced matrix from player B's point of view, the first column is dominated by the third column. Thus, by deleting the first row and then the first column, the reduced payoff matrix so obtained is:

	Player B		
Player A	B_2	B_3	B_4
A_2	4	2	4
A_3	2	4	0
A_4	4	0	8

Now it may be noted that none of the pure strategies of players A and B is inferior to any of their other strategies. However, the average of payoffs due to strategies B_3 and B_4, $\{(2 + 4)/2; (4 + 0)/2; (0 + 8)/2\} = (3, 2, 4)$ is superior to the payoff due to strategy B_2 of player B. Thus, strategy B_2 may be deleted from the matrix. The new matrix so obtained is:

	Player B	
Player A	B_3	B_4
A_2	2	4
A_3	4	0
A_4	0	8

Again in the reduced matrix, the average of the payoffs due to strategies A_3 and A_4 of player A, i.e. $\{(4 + 0)/2; (0 + 8)/2\} = (2, 4)$ is the same as the payoff due to strategy A_2. Therefore, player A will gain the same amount even if the strategy A_2 is never used. Hence, after deleting the strategy A_2 from the reduced matrix the following new reduced 2×2 payoff is obtained:

	Player B	
Player A	B_3	B_4
A_3	4	0
A_4	0	8

This game has no saddle point. Let player A choose his strategies A_3 and A_4 with probability p_1 and p_2, respectively, such that $p_1 + p_2 = 1$. Also let player B choose his strategies with probability q_1 and q_2, respectively, such that $q_1 + q_2 = 1$. Since both players want to retain their interests unchanged, therefore, we may write:

$$4p_1 + 0.p_2 = 0.p_1 + 8p_2 \qquad 4q_1 + 0.q_2 = 0.q_1 + 8q_2$$
$$4p_1 = 8(1 - p_1) \qquad 4q_1 = 8\,(1 - q_1)$$
$$p_1 = 2/3 \qquad q_1 = 2/3$$

We find that the optimal strategies of player A and player B in the original game are (0, 0, 2/3, 1/3) and (0, 0, 2/3, 1/3), respectively. The value of the game can be obtained by putting value of p_1 or q_1 in either of the expected payoff equations above. That is:

Expected gain to A	Expected loss to B
$4p_1 + 0.p_2 = 4(2/3) = 8/3$	$4q_1 + 0q_2 = 4(2/3) = 8/3$

Example 12.9 In a small town, there are only two stores, ABC and XYZ, that handle sundry goods. The total number of customers is equally divided between the two, because the price and the quality of goods sold are equal. Both stores have good reputation in the community, and they render equally good customer service. Assume that a gain of customers by ABC is a loss to XYZ and vice versa. Both stores plan to run annual pre-Diwali sales during the first week of November. Sales are advertised through a local newspaper, and through radio and television. With the aid of an advertising firm store, ABC constructed the game matrix given below. (Figures in the matrix represent a gain or loss of customers).

	Strategy of *XYZ*		
Strategy of *ABC*	*Newspaper*	*Radio*	*Television*
Newspaper	30	40	– 80
Radio	0	15	– 20
Television	90	20	50

Determine the optimal strategies and the worth of such strategies for both ABC and XYZ.

[*Delhi MBA, 2000, 2003; Jammu MBA, 2002*]

Solution The first step is to search for a saddle point. There is no saddle point in the problem.

The second step is to observe whether the payoff matrix can be reduced by rules of dominance. Yes, each element in first column is more than the corresponding element in the third column. Thus, for store XYZ, the newspaper as a media of advertisement is less attractive as compared to television. Therefore we remove column 1 from the payoff matrix. We now get:

	XYZ	
ABC	B_2	B_3
A_1	40	– 80
A_2	15	– 20
A_3	20	50

In the reduced payoff matrix, each element in the second row is less than the corresponding element in third row. So, store ABC will never like to choose radio as a media of advertisement because it would be earning less profit than it would if it advertised through television. Thus, by deleting the second row from the reduced matrix, we get the further reduced 2 × 2 payoff matrix as shown below:

	XYZ		
ABC	B_2	B_3	*Probability*
A_1	40	– 80	p_1
A_3	20	50	p_2
Probability	q_1	q_2	

Even the reduced 2 × 2 payoff matrix does not have the saddle point. Thus, both the stores use mixed strategies.

For Store ABC : Let p_1 and p_2 be probabilities of selecting strategy A_1 (newspaper) and A_3 (television), respectively. Then, the expected gain to store ABC when store XYZ uses its B_2 and B_3 strategies is given by:

$$40p_1 + 20p_2 \quad \text{and} \quad -80p_1 + 50p_2\ ; \quad p_1 + p_2 = 1$$

For store ABC the probability p_1 and p_2 should be such that the expected gains under both conditions are equal. That is

$$\begin{aligned} 40p_1 + 20p_2 &= -80p_1 + 50p_2 \\ 40p_1 + 20(1 - p_1) &= -80p_1 + 50(1 - p_1), \text{ where } p_1 + p_2 = 1 \\ 150p_1 &= 30 \text{ or } p_1 = 1/5, \text{ and } p_2 = 1 - p_1 = 4/5 \end{aligned}$$

Thus, the store ABC should apply strategy A_1 (newspaper) with a probability of 1/5 and strategy A_3 (television), with a probability of 4/5.

For Store XYZ : Let q_1 and q_2 be the probabilities of selecting strategy B_2 (radio) and B_3 (television), respectively. Then, the expected loss to store XYZ, when store ABC uses its strategies A_1 and A_3, should be

$$\begin{aligned} 40q_1 - 80q_2 &= 20q_1 + 50q_2 \text{ where } q_1 + q_2 = 1 \\ 40q_1 - 80(1 - q_1) &= 20q_1 + 50(1 - q_1) \\ 150q_1 &= 130 \quad \text{or} \quad q_1 = 13/15, \quad \text{and} \quad q_2 = 1 - q_1 = 2/15. \end{aligned}$$

Thus, store XYZ should apply strategy B_2 (radio) with a probability of 13/15 and strategy B_3 (television) with a probability of 2/15.

Substituting the values of p_1, p_2 or q_1, q_2 in any of the gain or loss equations, we shall get the expected value of the game (i.e. 24). This is shown below:

Expected Gain to Store ABC

(i) $40p_1 + 20p_2 = 40 \times (1/5) + 20 \times (4/5) = 24$

(ii) $-80p_1 + 50p_2 = -80 \times (1/5) + 50 \times (4/5) = 24$

Expected Loss to Store XYZ

(i) $40q_1 - 80q_2 = 40 \times (13/15) - 80 \times (2/15) = 24$

(ii) $20q_1 + 50q_2 = 20 \times (13/15) + 20 \times (2/15) = 24$

Here it may be noted that the expected loss to one store is the same as the expected gain to the other store.

Example 12.10 Two breakfast food manufacturers, ABC and XYZ are competing for an increased market share. The payoff matrix, shown in the following table, describes the increase in market share for ABC and decrease in market share of XYZ.

		XYZ		
ABC	*Give Coupons*	*Decrease Price*	*Maintain Present Strategy*	*Increase Advertising*
Give Coupons	2	– 2	4	1
Decrease Price	6	1	12	3
Maintain Present Strategy	– 3	2	0	6
Increase Advertising	2	– 3	7	1

Determine the optimal strategies for both the manufacturers and the value of the game.

[*Delhi Univ., MBA, Dec. 2004*]

Solution The first step is to search for a saddle point. There is no saddle point in the problem.

The second step is to observe if the payoff marix can be reduced in size by rules of dominance. Yes, each element of first row is less than the corresponding elements of second row. Therefore, the first row is dominated by second row because payoffs are less attractive for ABC. Thus, after deleting first row, the reduced matrix becomes as shown below:

		XYZ		
ABC	B_1	B_2	B_3	B_4
A_2	6	1	12	3
A_3	– 3	2	0	6
A_4	2	– 3	7	1

In the reduced matrix, each element of the fourth column is more than the corresponding element in second column. Therefore, the fourth column is dominated by the second column because the payoffs are less attractive (more loss) for XYZ. Thus, after deleting the fourth column the reduced matrix becomes:

	XYZ		
ABC	B_1	B_2	B_3
A_2	6	1	12
A_3	– 3	2	0
A_4	2	– 3	7

Further, compare rows 1 and 3 and then columns 1 and 3 and delete the less attractive row and column from ABC's and XYZ's point of view. The reduced payoff matrix is shown below:

	XYZ		
	Give Coupons	*Decrease Price*	
ABC	B_1	B_2	*Probability*
Decrease Price, A_2	6	1	p_1
Maintain Present Strategy, A_3	– 3	2	p_2
Probability	q_1	q_2	

Even the reduced 2 × 2 payoff matrix does not have the saddle point. Thus, both ABC and XYZ use mixed strategies.

For ABC : Let p_1 and p_2 be probabilities of selecting strategy A_2 (decrease price) and A_3 (maintain present strategy), respectively. Then the expected gain to ABC, when XYZ uses its B_1 and B_2 strategies, is given by:

$$6p_1 - 3p_2 \quad \text{and} \quad p_1 + 2\,p_2\,; \quad p_1 + p_2 = 1$$

The probability p_1 and p_2 should be such that the expected gains under both conditions are equal. That is:

$$6p_1 - 3p_2 = p_1 + 2p_2$$
$$6p_1 - 3(1 - p_1) = p_1 + 2(1 - p_1);\ p_1 + p_2 = 1$$
$$10p_1 = 5 \quad \text{or} \quad p_1 = 1/2 \quad \text{and} \quad p_2 = 1 - p_1 = 1/2$$

For XYZ : Let q_1 and q_2 be the probabilities of selecting strategies B_1 (give coupons) and B_2 (decrease price), respectively. Then, the expected loss to XYZ, when ABC uses its A_2 and A_3 strategies, should be:

$$6q_1 + q_2 = -3q_1 + 2q_2, \text{ where } q_1 + q_2 = 1$$
$$6q_1 + (1 - q_1) = -3q_1 + 2(1 - q_1)$$
$$10q_1 = 1 \quad \text{or} \quad q_1 = 1/10 \quad \text{and} \quad q_2 = 1 - q_1 = 9/10$$

Hence, the optimal strategies for both the manufacturers are that ABC should adopt strategy A_2 (decrease price) 50 per cent of time and strategy A_3 (maintain present strategy) 50 per cent of time, while XYZ should adopt strategy B_1 (give coupons) 10 per cent of time and strategy B_2 (decrease price), 90 per cent of time.

The expected gain and loss to ABC and XYZ can be calculated as shown below:

Expected gain to ABC:
$$6p_1 - 3p_2 = 6(1/2) - 3(1/2) = 3/2$$
$$p_1 + 2p_2 = 1/2 + 2(1/2) = 3/2$$

Expected loss to XYZ:
$$6q_1 + q_2 = 6(1/10) + (9/10) = 3/2$$
$$-3q_1 + 2q_2 = -3(1/10) + 2(9/10) = 3/2$$

Example 12.11 Two Firms A and B have, for years, been selling a competing product that forms a part of both firms' total sales. The marketing executive of Firm A raised the question: 'What should be the firm's strategies in terms of advertising for the product in question'. The market research team of Firm A developed the following data for varying degree of advertising:

(i) No advertising, medium advertising, and large advertising for both firms will result in equal market shares.

(ii) Firm *A* with no advertising: 40 per cent of the market with medium advertising by Firm *B* and 28 per cent of the market with large advertising by Firm *B*.

(iii) Firm *A* using medium advertising: 70 per cent of the market with no advertising by Firm *B* and 45 per cent of the market with large advertising by Firm *B*.

(iv) Firm *A* using large advertising: 75 per cent of the market with no advertising by Firm *B* and 47.5 per cent of the market with medium advertising by Firm *B*.

Based upon the above information, answer the marketing executive's questions.

What advertising policy should Firm *A* pursue when consideration is given to the above factors: selling price, Rs 4.00 per unit; variable cost of product, Rs 2.50 per unit; annual volume of 30,000 units for Firm *A*; cost of annual medium advertising Rs 5,000 and cost of annual large advertising Rs 15,000? What contribution, before other fixed costs, is available to the firm? [*Delhi Univ., MBA, 2001, 2005*]

Solution The payoff matrix of the game between Firms *A* and *B* is as follows:

		Firm *B*			
Firm *A*		*No Advt.*, B_1	*Medium Advt.*, B_2	*Large Advt.*, B_3	Row Minimum
No Advt.,	A_1	50	40	28	28
Medium Advt.,	A_2	70	50	45	45
Large Advt.,	A_3	75	47.5	50	47.5
Column Maximum		75	50	50	

From the payoff matrix it is observed that there is no saddle point in the problem.

Applying rules of dominance, delete the first row (dominated by third row) and then the first column (dominated by both columns 2 and 3) from the payoff matrix. The reduced payoff matrix, so obtained, is shown below:

	Firm *B*		
Firm *A*	B_2	B_3	Probability
A_2	50	45	p_1
A_3	47.5	50	p_2
Probability	q_1	q_2	

Even the reduced 2 × 2 payoff matrix does not have the saddle point. Thus, both the firms use mixed strategies. Adopting the same procedure, as discussed in earlier examples, the expected gain to Firm *A* can be calculated as follows:

Expected Gain to Firm A

$$50p_1 + 47.5p_2 = 45p_1 + 50p_2; \quad p_1 + p_2 = 1$$
$$50p_1 + 47.5(1 - p_1) = 45p_1 + 50(1 - p_1)$$
$$7.5p_1 = 2.5 \quad \text{or} \quad p_1 = 1/3 \quad \text{and} \quad p_2 = 1 - p_1 = 2/3.$$

$$\text{Expected gain} = 50p_1 + 47.5p_2$$
$$= 50(1/3) + 47.5(2/3) = 145/3.$$

Thus, the optimal policy for Firm *A* is to apply strategy A_2 (medium advertising) with probability 0.33 and strategy A_3 (large advertising) with probability 0.67 on any one play of the game. With this policy, the firm may expect to gain 145/3 = 48.3 per cent of the market share.

Table 12.5 Market Share of Film A

	Firm *B*		
Firm *A*	*No Advt.*	*Medium Advt.*	*Large Advt.*
No Advt.	0.50 × 30,000 = 15,000	0.40 × 30,000 = 12,000	0.28 × 30,000 = 8,400
Medium Advt.	0.70 × 30,000 = 21,000	0.50 × 30,000 = 15,000	0.45 × 30,000 = 13,500
Large Advt.	0.75 × 30,000 = 22,500	0.475 × 30,000 = 14,250	0.50 × 30,000 = 15,000

Given that the expenditure on medium and large advertisements is Rs 5,000 and Rs 15,000, respectively, net profit to Firm *A* can be calculated by using following equation:

Net profit = (Sales price – Cost price) × Sales volume – Advertising expenditure

The net profit to Firm *A* is shown in Table 12.6.

Table 12.6 Profit to Firm A

	Firm *B*		
Firm *A*	*No Advt.*	*Medium Advt.*	*Large Advt.*
No advt.	1.5 × 15,000 = 22,500	1.5 ×12,000 = 18,000	1.5 × 8,400 = 12,600
Medium advt.	1.5 × 21,000 – 5,000 = 26,500	1.5 × 15,000 – 5,000 = 17,500	1.5 × 13,500 – 5,000 = 15,250
Large advt.	1.5 × 22,500 – 15,000 = 18,750	1.5 × 14,250 – 15,000 = 6,375	1.5 × 15,000 – 15,000 = 7,500

Observations

1. If Firm *A* chooses the strategy of *No advertising*, then the minimum profit is Rs 12,600, because Firm *B* can adopt its strategy *Large advertising*.
2. If Firm *A* chooses the strategy of *Medium advertising*, then the minimum profit is Rs 15,250 because Firm *B* can again adopt its strategy *Large advertising*.
3. If Firm *A* chooses the strategy of *Large advertising*, then the minimum profit is Rs 6,375 because firm *B* can adopt its strategy *Medium advertising*.

Based on these observations, the Firm *A* must adopt the policy of 'Medium Advertising' in order to gain the maximum profit of Rs 15,250 among these three alternatives and must spend Rs 5,000 for advertising.

12.6.2 Arithmetic Method

The arithmetic method (also known as *short-cut method*) provides an easy method for finding optimal strategies for each player in a payoff matrix of size 2 × 2, without saddle point. The steps of this method are as follows:

1. Find the differences between the two values in the first row and put it against the second row of the matrix, neglecting the negative sign (if any).

2. Find the difference between the two values in the second row and put it against first row of the matrix, neglecting the negative sign (if any).
3. Repeat Steps 1 and 2 for the two columns also.

The values obtained by 'swapping the differences' represent the optimal relative frequencies of play for both players strategies. These may be converted to probabilities by dividing each of them by their sum.

The value of the game can be obtained by applying any of the methods discussed earlier.

Remark The arithmetic method should not be used to solve a 2 × 2 game that has a saddle point because the method yields an incorrect answer.

Example 12.12 Two competitors are competing for the market share of the similar product. The payoff matrix in terms of their advertising plan is shown below:

	Competitor *B*		
Competitor *A*	*No Advertising*	*Medium Advertising*	*Heavy Advertising*
No Advertising	10	5	–2
Medium Advertising	13	12	13
Heavy Advertising	16	14	10

Suggest optimal strategies for the two firms and the net outcome thereof.

[*Delhi Univ., MCom, 2000*]

Solution Applying rules of dominance to delete first column (dominated by second column) and then first row (dominated by second as well as third rows) from the payoff matrix, we obtain the following reduced payoff matrix:

	Firm *B*	
Firm *A*	*Medium Advt.* B_2	*Heavy Advt.* B_3
Medium Advt. A_2	12	15
Heavy Advt. A_3	14	10

As the payoff matrix does not have a saddle point, firms will use mixed strategies. Applying arithmetic method as explained earlier to get optimal mixed strategies for both the firms, the results are:

	Firm *B*	
Firm *A*	B_2	B_3
A_2	12	15
A_3	14	10

$14 - 10 = 4$, $p(A_2) = \dfrac{4}{4+3} = \dfrac{4}{7}$

$15 - 12 = 3$, $p(A_3) = \dfrac{3}{4+3} = \dfrac{3}{7}$

$15 - 10 = 5$ $\qquad$ $14 - 12 = 2$

$$p(B_2) = \frac{5}{5+2} = \frac{5}{7} \qquad p(B_3) = \frac{2}{5+2} = \frac{2}{7}$$

Hence, Firm *A* should adopt strategy A_2 and A_3, 57 per cent of the time and 43 per cent of time, respectively (or with 57 per cent and 43 per cent probability on any one play of the game, respectively). Similarly, Firm *B* should adopt strategy B_2 and B_3, 71 per cent of time and 29 per cent of time, respectively (or with 71 per cent and 29 per cent probability on any one play of the game, respectively).

Expected Gain to Firm A

(i) $12 \times (4/7) + 14 \times (3/7) = 90/7$, Firm *B* adopt B_2
(ii) $15 \times (4/7) + 10 \times (3/7) = 90/7$, Firm *B* adopt B_3

Expected Loss to Firm B

(i) $12 \times (5/7) + 15 \times (2/7) = 90/7$, Firm *A* adopt A_2
(ii) $14 \times (5/7) + 10 \times (2/7) = 90/7$, Firm *A* adopt A_3

12.6.3 Matrix Method

If the game matrix is in the form of a square matrix, then the optimal strategy mix as well as value of the game may be obtained by the matrix method. The solution of a two-person zero-sum game with mixed strategies with a square payoff matrix may be obtained by using the following formulae:

$$\text{Player } A\text{'s optimal strategy} = \frac{[1 \quad 1]\, P_{\text{adj}}}{[1 \quad 1]\, P_{\text{adj}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

$$\text{Player } B\text{'s optimal strategy} = \frac{[1 \quad 1]\, P_{\text{cof}}}{[1 \quad 1]\, P_{\text{adj}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

Value of the game = (Player A's optimal strategies) × (Payoff matrix p_{ij}) × (Player B's optimal strategies) where P_{adj} = adjoint matrix, P_{cof} = cofactor matrix. Player A's optimal strategies are in the form of a row vector and B's optimal strategies are in the form of a column vector.

This method can be used to find a solution of a game with size of more than 2 × 2. However, in rare cases, the solution violates the non-negative condition of probabilities, i.e. $p_i \geq 0$, $q_j \geq 0$, although the requirement $p_1 + p_2 + \ldots + p_m = 1$ or $q_1 + q_2 + \ldots + q_n = 1$ is met.

Example 12.13 Solve the following game after reducing it to a 2 × 2 game

	Player B		
Player A	B_1	B_2	B_3
A_1	1	7	2
A_2	6	2	7
A_3	5	1	6

Solution In the given game matrix, the third row is dominated by the second row and in the reduced matrix third column is dominated by the first column. So, after elimination of the third row and the third column the game matrix becomes:

	Player B	
Player A	B_1	B_2
A_1	1	7
A_2	6	2

For this reduced matrix, let us calculate P_{adj} and P_{cof} as given below:

$$P_{\text{adj}} = \begin{bmatrix} 2 & -7 \\ -6 & 1 \end{bmatrix} \quad \text{and} \quad P_{\text{cof}} = \begin{bmatrix} 2 & -6 \\ -7 & 1 \end{bmatrix}$$

$$\text{Player } A\text{'s optimal strategies} = \frac{[1 \quad 1]\begin{bmatrix} 2 & -7 \\ -6 & 1 \end{bmatrix}}{[1 \quad 1]\begin{bmatrix} 2 & -7 \\ -6 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}} = \frac{[-4 \quad -6]}{-10} = \frac{[4 \quad 6]}{10}$$

This solution can be broken down into the optimal strategy mix for player A as $p_1 = 4/10 = 2/5$ and $p_2 = 6/10 = 3/5$, where p_1 and p_2 represent the probabilities of player A's using his strategies A_1 and A_2, respectively.

Similarly, the optimal strategy mixture for player B is obtained as:

$$\text{Player } B\text{'s optimal strategies} = \frac{[1 \quad 1]\begin{bmatrix} 2 & -6 \\ -7 & 1 \end{bmatrix}}{[1 \quad 1]\begin{bmatrix} 2 & -7 \\ -6 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}} = \frac{[-5 \quad -5]}{-10} = \frac{[5 \quad 5]}{10}$$

This solution can also be broken down into the optimal strategy mixture for player B as $q_1 = 5/10 = 1/2$ and $q_2 = 5/10 = 1/2$, where q_1 and q_2 represent the probabilities of player B's using his strategies B_1 and B_2, respectively. Hence:

Value of the game, $V = \begin{bmatrix} \frac{2}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 & 7 \\ 6 & 2 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = 4$

12.6.4 Graphical Method

The graphical method is useful for the game where the payoff matrix is of the size $2 \times n$ or $m \times 2$, i.e. the game with mixed strategies that has only two undominated pure strategies for one of the players in the two-person zero-sum game.

Optimal strategies for both the players assign non-zero probabilities to the same number of pure strategies. Therefore, if one player has only two strategies, the other will also use the same number of strategies. Hence, this method is useful in finding out which of the two strategies can be used.

Consider the following $2 \times n$ payoff matrix of a game, without saddle point.

	Player B				
Player A	B_1	B_2	...	B_n	*Probability*
A_1	a_{11}	a_{12}	...	a_{1n}	p_1
A_2	a_{21}	a_{22}	...	a_{2n}	p_2
Probability	q_1	q_2	...	q_n	

Player A has two strategies A_1 and A_2 with probability of their selection p_1 and p_2, respectively, such that $p_1 + p_2 = 1$ and $p_1, p_2 \geq 0$. Now for each of the pure strategies available to player B, the expected pay off for player A would be as follows:

B's Pure Strategies	*A's Expected Payoff*
B_1	$a_{11}p_1 + a_{21}\ p_2$
B_2	$a_{12}p_1 + a_{22}\ p_2$
⋮	⋮
B_n	$a_{1n}p_1 + a_{2n}\ p_2$

According to the maximin criterion for mixed strategy games, player A should select the value of probability p_1 and p_2 so as to maximize his minimum expected payoffs. This may be done by plotting the straight lines representing player A's expected payoff values.

The highest point on the lower boundary of these lines will give the maximum expected payoff among the minimum expected payoffs and the optimum value of probability p_1 and p_2.

Now, the two strategies of player B corresponding to those lines which pass through the maximin point can be determined. This helps in reducing the size of the game to (2×2), which can be easily solved by any of the methods discussed earlier.

The $(m \times 2)$ games are also treated in the same way except that the upper boundary of the straight lines corresponding to B's expected payoff will give the maximum expected payoff to player B and the lowest point on this boundary will then give the minimum expected payoff (minimax value) and the optimum value of probability q_1 and q_2.

Example 12.14 Use the graphical method for solving the following game and find the value of the game.

	Player B			
Player A	B_1	B_2	B_3	B_4
A_1	2	2	3	−2
A_2	4	3	2	6

Solution The game does not have a saddle point. If the probability of player A's playing A_1 and A_2 in the strategy mixture is denoted by p_1 and p_2, respectively, where $p_2 = 1 - p_1$, then the expected payoff (gain) to player A will be

B's Pure Strategies	*A's Expected Payoff*
B_1	$2\ p_1 + 4\ p_2$
B_2	$2\ p_1 + 3\ p_2$
B_3	$3\ p_1 + 2\ p_2$
B_4	$-2\ p_1 + 6\ p_2$

These four expected payoff lines can be plotted on a graph to solve the game.

The graph for player A A graphic solution is shown in Fig. 12.2. Here, the probability of player A's playing A_1, i.e. p_1 is measured on the x-axis. Since p_1 cannot exceed 1, the x-axis is cutoff at $p_1 = 1$. The expected payoff of player A is measured along y-axis. From the game matrix, if player B plays B_1, the expected payoff of player A is 2 when A plays A_1 with $p_1 = 1$ and 4 when A plays A_2 with $p_1 = 0$. These two extreme points are connected by a straight line, which shows the expected payoff of A when B plays B_1. Three other straight lines are similarly drawn for B_2, B_3 and B_4.

It is assumed that player B will always play his best possible strategies yielding the worst result to player A. Thus, the payoffs (gains) to A are represented by the lower boundary when he is faced with the most unfavourable situation in the game. Since player A must choose his best possible strategies in order to realize a maximum expected gain, the highest expected gain is found at point P, where the two straight lines

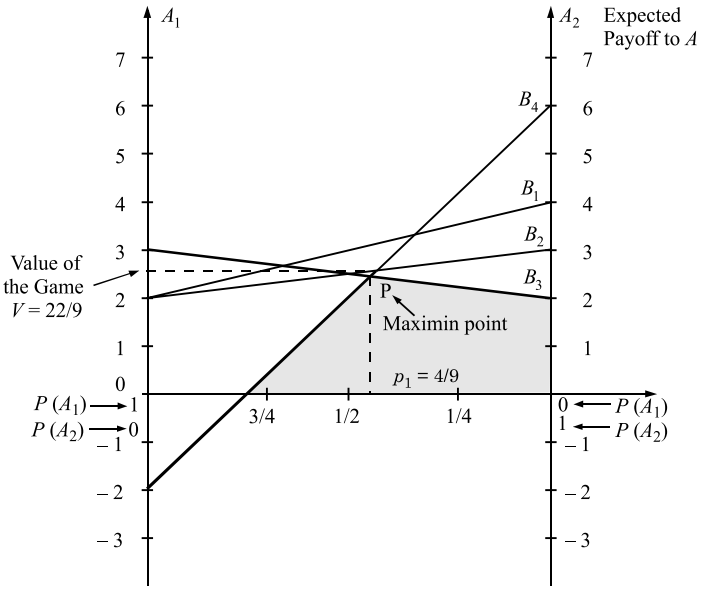


Fig. 12.2 Graph for Player *A*

$$E_3 = \quad 3p_1 + 2p_2 = \quad 3p_1 + 2(1 - p_1)$$
$$E_4 = -\ 2p_1 + 6p_2 = -\ 2p_1 + 6(1 - p_1)$$

meet. In this manner the solution to the original (2×4) game reduces to that of the game with payoff matrix of size (2×2) as given below:

	Player B	
Player A	B_3	B_4
A_1	3	-2
A_2	2	6

The optimum payoff to player A can now be obtained by setting E_3 and E_4 equal and solving for p_1, i.e.

$$3p_1 + 2(1 - p_1) = -\ 2p_1 + 6(1 - p_1) \quad \text{or} \quad p_1 = 4/9; \quad p_2 = 1 - p_1 = 5/9$$

Substituting the value of p_1 and p_2 in the equation for E_3 (or E_4) we have:

$$\text{Value of the game, } V = 3 \times 4/9 + 2 \times 5/9 = 22/9$$

The optimal strategy mix of player B can also be found in the same manner as for player A. If the probabilities of B's selecting strategy B_3 and B_4 are denoted by q_3 and q_4, respectively, then the expected loss to B will be:

$$L_3 = 3q_3 - 2q_4 = 3q_3 - 2(1 - q_3) \quad (\text{if } A \text{ selects } A_1)$$
$$L_4 = 2q_3 + 6q_4 = 2q_3 + 6(1 - q_3) \quad (\text{if } A \text{ selects } A_2)$$

To solve for q_3, equate the two equations:

$$3q_3 - 2(1 - q_3) = 2q_3 + 6(1 - q_3) \quad \text{or} \quad q_3 = 8/9;\ q_4 = 1 - q_3 = 1/9$$

Substituting the value of q_3 and q_4 in the equation for L_3 (or L_4), we have

Value of the game, $V = 3 \times 8/9 - 2 \times 1/9 = 22/9$

Example 12.15 Two firms A and B make colour and black & white television sets. Firm A can make either 150 colour sets in a week or an equal number of black & white sets, and make a profit of Rs 400 per colour set, or 150 colour and 150 black & white sets, or 300 black & white sets per week. It also has the same profit margin on the two sets as A. Each week there is a market of 150 colour sets and 300 black & white sets and the manufacturers would share market in the proportion in which they manufacture a particular type of set.

Write the pay-off matrix of A per week. Obtain graphically *A's* and *B's* optimum strategies and value of the game. [*Bombay Univ.*, *MMS*, *2000*]

Solution For firm A, the strategies are:

A_1 : make 150 colour sets, A_2 : make 150 black & white sets.

For firm B, the strategies are:

B_1 : make 300 colour sets, B_2 : make 150 colour and 150 black & white sets.

B_3 : make 300 black and white sets.

For the combination A_1B_1, the profit to firm A would be: $\{150/(150+300)\} \times 150 \times 400 = \text{Rs } 20{,}000$ wherein $150/(150+300)$ represents share of market for A, 150 is the total market for colour television sets and 400 is the profit per set. In a similar manner, other profit figures may be obtained as shown in the following pay-off matrix:

	B's Strategy		
A's Strategy	B_1	B_2	B_3
A_1	20,000	30,000	60,000
A_2	45,000	45,000	30,000

This pay-off table has no saddle point. Thus to determine optimum mixed strategy, the data are plotted on graph as shown in Fig. 12.3.

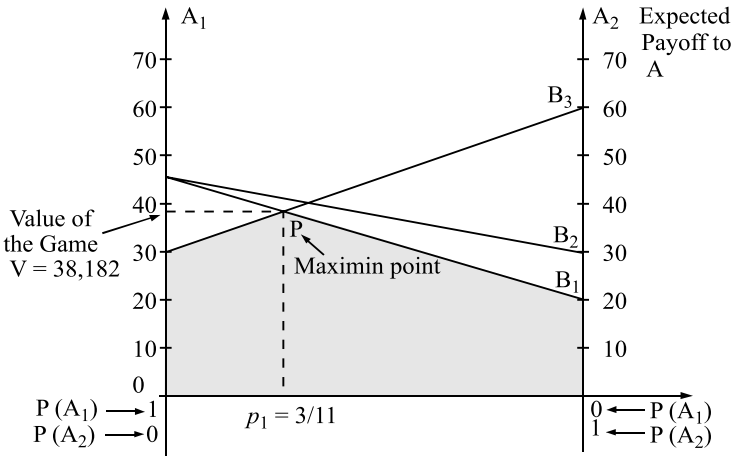


Fig. 12.3 Graph for Player *A*

Lines joining the pay-offs on axis A_1 with the pay-offs on axis A_2 represents each of *B's* strategies. Since firm A wishes to maximize his minimum expected pay-off, we consider the highest point of intersection, P on the lower envelope of *A's* expected pay-off equation. This point P represents the maximum expected value of the game. The lines B_1 and B_3 passing through P, define the strategies which firm B needs to adopt. The solution to the original 2×3 game, therefore, reduces to that of the simpler game with 2×2 pay-off matrix as follows:

	B's Strategy		
A's Strategy	B_1	B_3	Probability
A_1	20,000	60,000	p_1
A_2	45,000	30,000	p_2
Probability	q_1	q_2	

The optimal mixed strategies of player A are: $A_1 = 3/11, A_2 = 8/11$. Similarly, the optimal mixed strategies for B are: $B_1 = 6/11, B_2 = 0, B_3 = 5/11$. The value of the game is $V = 38{,}182$.

Example 12.16 Obtain the optimal strategies for both persons and the value of the game for two-person zero-sum game whose payoff matrix is as follows:

	Player B	
Player A	B_1	B_2
A_1	1	– 3
A_2	3	5
A_3	– 1	6
A_4	4	1
A_5	2	2
A_6	– 5	0

Solution The game does not have any saddle point. If the probability of player B's playing strategies B_1 and B_2 in the strategy mix is denoted by q_1 and q_2 such that $q_1 + q_2 = 1$, then the expected payoff to player B will be:

A's Pure Strategies	*B's Expected Payoff*
A_1	$q_1 - 3q_2$
A_2	$3q_1 + 5q_2$
A_3	$- q_1 + 6q_2$
A_4	$4q_1 + q_2$
A_5	$2q_1 + 2q_2$
A_6	$- 5q_1 + 0q_2$

The six expected payoff lines can be plotted on the graph to solve the game.

The graph for player B A graphic solution is shown in Fig. 12.3 where the probability of player B's playing B_1, i.e. q_1 is measured on the x-axis. Since q_1 cannot exceed 1, therefore the x-axis is cutoff at $q_1 = 1$. The expected payoff of player B is measured along y-axis. From the game matrix, if player A plays A_1, the expected payoff of player B is 1 when he plays B_1 with $q_1 = 1$ and –3 when he plays B_2 with $q_1 = 0$. These two extreme points are connected by a straight line, which shows the expected payoff to B when A plays A_1. Five other straight lines are similarly drawn for A_2 to A_6.

It is assumed that player A will always play his best possible strategies, yielding the worst result to player B. Thus, payoffs (losses) to B are represented by the upper boundary when he is faced with the most unfavourable situation in the game. According to the minimax criterion, player B will always select a combination of strategies B_1 and B_2, so that he minimizes the losses. Even in this case the optimum solution occurs at the intersection of the two payoff lines.

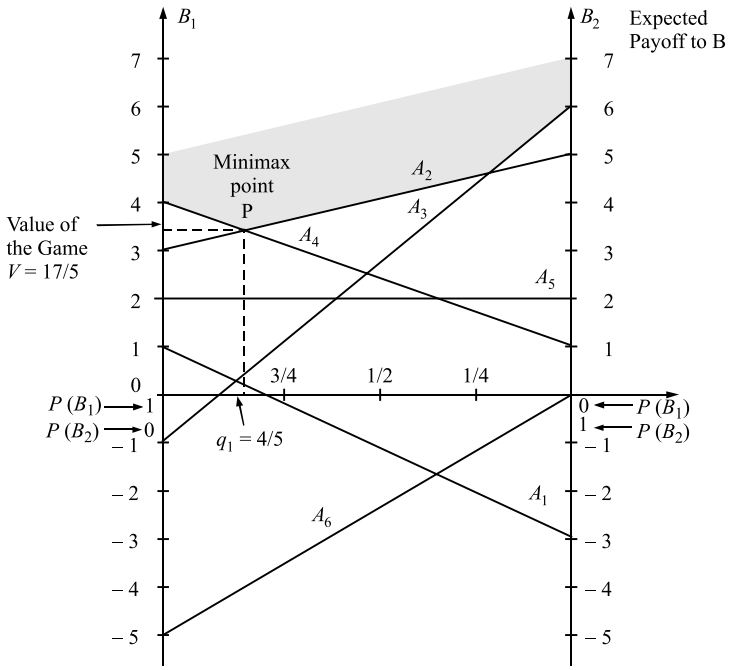


Fig. 12.4 Graph for Player *B*

$$E_3 = 3q_1 + 5q_2 = 3q_1 + 5(1 - q_1)$$
$$E_4 = 4q_1 + \ q_2 = 4q_1 + (1 - q_1)$$

The solution to the original (6 × 2) game reduces to that of the game with payoff matrix of size (2 × 2) as shown below:

	Player B	
Player A	B_1	B_2
A_2	3	5
A_4	4	1

Now using the usual method of solution for a (2 × 2) game, the optimum strategies can be obtained as given below:

Player A: (0, 3/5, 0, 2/5, 0, 0) ; Player B: (4/5, 1/5)

and, Value of the game, $V = 17/5$.

Example 12.17 Solve the following game graphically:

	Player B	
Player A	B_1	B_2
A_1	1	2
A_2	4	5
A_3	9	– 7
A_4	– 3	– 4
A_5	2	1

[*Madurai Univ.*, *M Sc* (*Maths*), *2002*]

Solution The given pay-off matrix has no saddle point. So, let the player B play the mixed strategies with probability q_1 and q_2 such that, $q_1 + q_2 = 1$. Then, B's expected pay-offs against *A's* pure strategies are given by

A's Pure Action	*B's Expected Pay-off* (E_i)	
A_1	q_1	$+ 2q_2$
A_2	$4q_1$	$+ 5q_2$
A_3	$9q_1$	$+ 7q_2$
A_4	$- 3q_1$	$- 4q_2$
A_5	$2q_1$	$+ \ q_2$

The graph for player B: The expected pay-off equations are plotted as shown in Fig. 12.5. The point P represents the minimax expected value of the game for player B. The minimal point occurs at the intersection of two pay-off lines

$$E_2 = 4q_1 + 5q_2$$

and $$E_3 = 9q_1 - 7q_2.$$

The solution to the 5 × 2 game reduces to that of the game with pay-off matrix of size (2 × 2). The optimum pay-off to player B can be obtained by setting E_2 and E_3 equal and solving for q_1, i.e.

$$4q_1 + 5q_2 = 4q_1 - 7q_2$$

or $$4q_1 + 5(1 - q_1) = 9q_1 - 7(1 - q_1)$$

or $$q_1 = 12/17 \text{ and}$$
$$q_2 = 1 - q_1 = 5/17.$$

Substituting the value of q_1 and q_2 in equation E_2 (or E_3), we have the expected loss to B as $V = 73/13$.

If the probability of player *A's* selecting strategy A_2 and A_3 are p_2 and p_3, respectively, then the expected gain to A can be calculated by equating two pay-off lines: $4p_2 + 9p_3 = 5p_2 - 7p_3$ to obtain $p_2 = 16/17, p_3 = 1 - p_2 = 1/17$ whereas $p_1 = 0$ and $p_4 = 0$. The expected gain to A is $V = 73/17$.

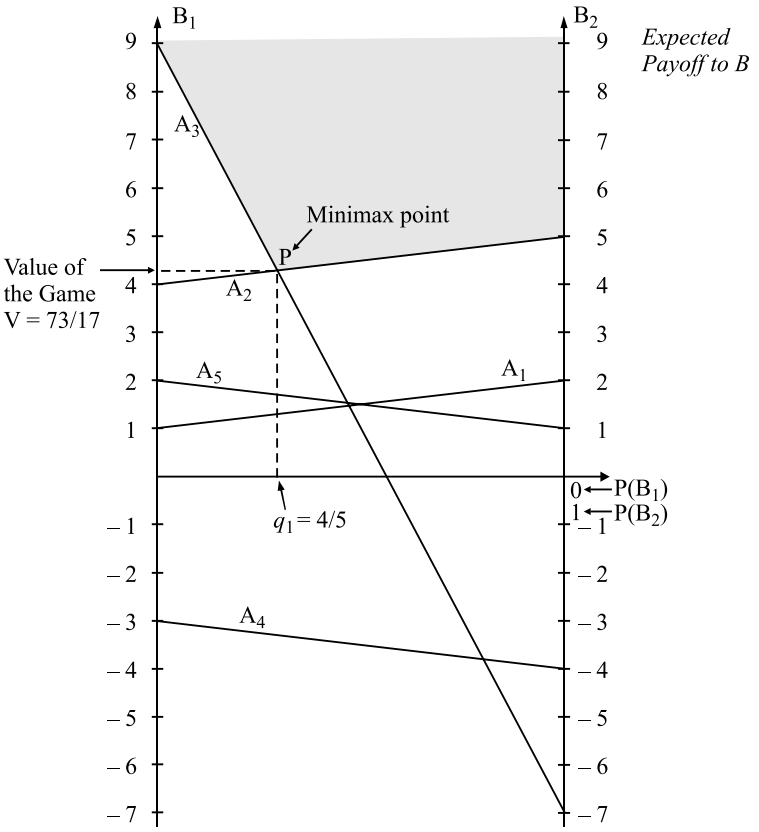


Fig. 12.5 Graph for Player *B*

12.6.5 Linear Programming Method

The two-person zero-sum games can also be solved by linear programming. The major advantage of using linear programming technique is that it helps to solve the mixed-strategy games of larger dimension payoff matrix.

To illustrate the transformation of a game problem to a linear programming problem, consider a payoff matrix of size $m \times n$. Let a_{ij} be the element in the ith row and jth column of game payoff matrix, and letting p_i be the probabilities of m strategies ($i = 1, 2, \ldots, m$) for player A. Then, the expected gains for player A, for each of player B's strategies will be:

$$V = \sum_{i=1}^{m} p_i \, a_{ij}, \qquad j = 1, 2, \ldots, n$$

The aim of player A is to select a set of strategies with probability p_i ($i = 1, 2, \ldots, m$) on any play of game such that he can maximize his minimum expected gains.

Now to obtain values of probability p_i, the value of the game to player A for all strategies by player B must be at least equal to V. Thus to maximize the minimum expected gains, it is necessary that:

$$\begin{aligned} a_{11}\, p_1 + a_{21}\, p_2 + \ldots + a_{m1}\, p_m &\geq V \\ a_{12}\, p_1 + a_{22}\, p_2 + \ldots + a_{m2}\, p_m &\geq V \\ \vdots \qquad\qquad\qquad & \vdots \\ a_{1n}\, p_1 + a_{2n}\, p_2 + \ldots + a_{mn}\, p_m &\geq V \end{aligned}$$

where $$p_1 + p_2 + \ldots + p_m = 1, \text{ and } p_i \geq 0 \quad \text{for all } i.$$

Dividing both sides of the m inequalities and equation by V the division is valid as long as $V > 0$. In case $V < 0$, the direction of inequality constraints must be reversed. But if $V = 0$, the division would be meaningless. In this case a constant can be added to all entries of the matrix, ensuring that the value of the game (V) for the revised matrix becomes more than zero. After optimal solution is obtained, the true value of the game is obtained by substracting the same constant value. Let $p_i/V = x_i$, (≥ 0), we then have

$$a_{11}\frac{p_1}{V} + a_{21}\frac{p_2}{V} + \ldots + a_{m1}\frac{p_m}{V} \geq 1$$
$$a_{12}\frac{p_1}{V} + a_{22}\frac{p_2}{V} + \ldots + a_{m2}\frac{p_m}{V} \geq 1$$
$$\vdots \qquad\qquad \vdots$$
$$a_{1n}\frac{p_1}{V} + a_{2n}\frac{p_2}{V} + \ldots + a_{mn}\frac{p_m}{V} \geq 1$$
$$\frac{p_1}{V} + \frac{p_2}{V} + \ldots + \frac{p_m}{V} = 1$$

Since the objective of player A is to maximize the value of the game, V, which is equivalent to minimizing $1/V$, the resulting linear programming problem can be stated as:

Minimize $Z_p\ (= 1/V) = x_1 + x_2 + \ldots + x_m$

subject to the constraints

$$a_{11}\,x_1 + a_{21}\,x_2 + \ldots + a_{m1}\,x_m \geq 1$$
$$a_{12}\,x_1 + a_{22}\,x_2 + \ldots + a_{m2}\,x_m \geq 1$$
$$\vdots \qquad\qquad \vdots$$
$$a_{1n}\,x_1 + a_{2n}\,x_2 + \ldots + a_{mn}\,x_m \geq 1$$

and $x_1, x_2, \ldots, x_m \geq 0$

where $x_i = p_i/V \geq 0;\ i = 1, 2, \ldots, m$

Similarly, Player B has a similar problem with the inequalities of the constraints reversed, i.e. minimize the expected loss. Since minimizing V is equivalent to maximizing $1/V$, therefore, the resulting linear programming problem can be stated as:

Maximize $Z_q\ (= 1/V) = y_1 + y_2 + \ldots + y_n$

subject to the constraints

$$a_{11}\,y_1 + a_{12}\,y_2 + \ldots + a_{1n}\,y_n \leq 1$$
$$a_{21}\,y_1 + a_{22}\,y_2 + \ldots + a_{2n}\,y_n \leq 1$$
$$\vdots \qquad\qquad \vdots$$
$$a_{m1}\,y_1 + a_{m2}\,y_2 + \ldots + a_{mn}\,y_n \leq 1$$
$$y_1, y_2, \ldots, y_n \geq 0$$

where, $y_j = q_j/V \geq 0;\quad j = 1, 2, \ldots, n$

It may be noted that the LP problem for player B is the dual of LP problem for player A and vice versa. Therefore, the solution of the dual problem can be obtained from the primal simplex table. Since for both the players $Z_p = Z_q$, the expected gain to player A in the game will be exactly equal to expected loss to player B.

Remark Linear programming technique requires all variables to be non-negative and therefore to obtain a non-negative of value V of the game, the data to the problem, i.e. a_{ij} in the payoff table should all be non-negative. If there are some negative elements in the payoff table, a constant to every element in the payoff table must be added so as to make the smallest element zero; the solution to this new game will give an optimal mixed strategy for the original game. The value of the original game then equals the value of the new game minus the constant.

Example 12.18 For the following payoff matrix, transform the zero-sum game into an equivalent linear programming problem and solve it by using the simplex method.

	Player B		
Player A	B_1	B_2	B_3
A_1	1	–1	3
A_2	3	5	–3
A_3	6	2	–2

Solution The first step is to find out the saddle point (if any) in the payoff matrix as shown below:

	Player B			
Player A	B_1	B_2	B_3	Row minimum
A_1	1	[–1]	(3)	[–1] ← *Maximin*
A_2	3	5	–3	–3
A_3	6	2	–2	–2
Column maximum	6	5	(3)	← *Minimax*

The given game payoff matrix does not have a saddle point. Since the maximin value is –1, therefore, it is possible that the value of game (V) may be negative or zero because $-1 < V < 1$. Thus, a constant that is at least equal to the negative of maximin value, i.e. more than –1 is added to all the elements of the payoff matrix. Thus, adding a constant number 4 to all the elements of the payoff matrix, the payoff matrix becomes:

	Player B			
Player A	B_1	B_2	B_3	Probability
A_1	5	3	7	p_1
A_2	7	9	1	p_2
A_3	10	6	2	p_3
Probability	q_1	q_2	q_3	

Let p_i (i = 1, 2, 3) and q_j (j = 1, 2, 3) be the probabilities of selecting strategies A_i (i = 1, 2, 3) and B_j (j = 1, 2, 3) by players A and B, respectively.

The expected gain for player A will be as follows:

$$5p_1 + 7p_2 + 10p_3 \geq V \quad \text{(if } B \text{ uses strategy } B_1\text{)}$$
$$3p_1 + 9p_2 + 6p_3 \geq V \quad \text{(if } B \text{ uses strategy } B_2\text{)}$$
$$7p_1 + p_2 + 2p_3 \geq V \quad \text{(if } B \text{ uses strategy } B_3\text{)}$$
$$p_1 + p_2 + p_3 = 1$$

and
$$p_1, p_2, p_3 \geq 0$$

Dividing each inequality and equality by V, we get,

$$5\frac{p_1}{V} + 7\frac{p_2}{V} + 10\frac{p_3}{V} \geq 1, \qquad 3\frac{p_1}{V} + 9\frac{p_2}{V} + 6\frac{p_3}{V} \geq 1$$
$$7\frac{p_1}{V} + \frac{p_2}{V} + 2\frac{p_3}{V} \geq 1, \qquad \frac{p_1}{V} + \frac{p_2}{V} + \frac{p_3}{V} = \frac{1}{V}$$

In order to simplify, we define new variables:

$$x_1 = p_1/V, \quad x_2 = p_2/V \quad \text{and} \quad x_3 = p_3/V$$

The problem for player A, therefore becomes,

Minimize Z_p (= $1/V$) = $x_1 + x_2 + x_3$

subject to the constraints

(i) $5x_1 + 7x_2 + 10x_3 \geq 1$, (ii) $3x_1 + 9x_2 + 6x_3 \geq 1$, (iii) $7x_1 + x_2 + 2x_3 \geq 1$

and $x_1, x_2, x_3 \geq 0$

Player B's objective is to minimize his expected losses that can be reduced to minimizing the value of the game V. Hence, the problem of player B can be expressed as follows:

Maximize Z_q (= $1/V$) = $y_1 + y_2 + y_3$

subject to the constraints

(i) $5y_1 + 3y_2 + 7y_3 \leq 1$, (ii) $7y_1 + 9y_2 + y_3 \leq 1$, (iii) $10y_1 + 6y_2 + 2y_3 \leq 1$

and $y_1, y_2, y_3 \geq 0$

where $y_1 = q_1/V$; $y_2 = q_2/V$ and $y_3 = q_3/V$.

It may be noted here the that problem of player A is the dual of the problem of player B. Therefore, the solution of the dual problem can be obtained from the optimal simplex table of primal.

To solve the problem of player B, introduce slack variables to convert the three inequalities to equalities. The problem now becomes:

Maximize $Z_q = y_1 + y_2 + y_3 + 0s_1 + 0s_2 + 0s_3$

subject to the constraints

(i) $5y_1 + 3y_2 + 7y_3 + s_1 = 1$, (ii) $7y_1 + 9y_2 + y_3 + s_2 = 1$

(iii) $10y_1 + 6y_2 + 2y_3 + s_3 = 1$

and $y_1, y_2, y_3, s_1, s_2, s_3 \geq 0$

The initial solution is shown in Table 12.7.

	$c_j \rightarrow$	*1*	*1*	*1*	*0*	*0*	*0*	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Solution* y_B (= b)	y_1	y_2	y_3	s_1	s_2	s_3	*Min. Ratio* y_B/y_1
0	s_1	1	5	3	7	1	0	0	1/5
0	s_2	1	7	9	1	0	1	0	1/7
0	s_3	1	(10)	6	2	0	0	1	1/10 →
$Z = 0$		z_j	0	0	0	0	0	0	
		$c_j - z_j$	1	1	1	0	0	0	

↑

Table 12.7 Initial Solution

Proceeding with the usual simplex method, the optimal solution is shown in Table 12.8.

	$c_j \rightarrow$	*1*	*1*	*1*	*0*	*0*	*0*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Solution* y_B (= b)	y_1	y_2	y_3	s_1	s_2	s_3
1	y_3	1/10	2/5	0	1	3/20	– 1/10	0
1	y_2	1/10	11/15	1	0	– 1/60	7/60	0
0	s_3	1/5	24/5	0	0	– 1/5	– 3/5	1
$Z = 1/5$		z_j	17/15	0	0	2/15	1/15	0
		$c_j - z_j$	– 2/15	0	0	– 2/15	– 1/15	0

Table 12.8 Optimal Solution

The optimal solution (mixed strategies) for B is: $y_1 = 0$; $y_2 = 1/10$ and $y_3 = 1/10$ and the expected value of the game is: $Z = 1/V$ – constraint (= 4) = 5 – 4 = 1.

These solution values are now converted back into the original variables: If $1/V = 1/5$ then $V = 5$

$y_1 = q_1/V$, then $q_1 = y_1 \times V = 0$

$y_2 = q_2/V$, then $q_2 = y_2 \times V = 1/10 \times 5 = 1/2$

$y_3 = q_3/V$, then $q_3 = y_3 \times V = 1/10 \times 5 = 1/2$

The optimal strategies for player A are obtained from the $c_j - z_j$ row of the Table 12.8.

$x_1 = 2/15$, $x_2 = 1/15$ and $x_3 = 0$

Then $p_1 = x_1 \times V = (2/15) \times 5 = 2/3$; $p_2 = x_2 \times V = (1/15) \times 5 = 1/3$; $p_3 = x_3 \times V = 0$

Hence, the probabilities of using strategies by both the players are:

Player A = (2/3, 1/3, 0); Player B = (0, 1/2, 1/2) and, Value of the game $V = 1$.

CONCEPTUAL QUESTIONS B

1. Explain the following terms:
   (i) Two-person zero-sum game,
   (ii) Principles of dominance,
   (iii) Pure strategy in game theory
2. How is the concept of dominance used in simplifying the solution of a rectangular game?
3. Explain the theory of dominance in the solution of rectangular games.
4. Explain the graphical method of solving $2 \times n$ and $m \times 2$ games.

SELF PRACTICE PROBLEMS B

1. Consider a modified form of 'matching coins' game problem. The matching player is paid Rs 8, if the two coins turn both heads and Re 1 if the coins turn both tails. The non-matching player is paid Rs 3 when the two coins do not match. Given the choice of being a matching or non-matching player, which one would you choose and what would be your strategy?
2. Use dominance rules to reduce the size of the following payoff matrix to (2 × 2) size and hence, find the optimal strategies and value of the game.

(a)

Player *A*	Player *B* B_1	B_2	B_3
A_1	3	– 2	4
A_2	– 1	4	2
A_3	2	2	6

[*Punjab MSc* (*Stat.*), *1995*]

(b)

Player *A*	Player *B* B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

3. Players *A* and *B*, each take out one or two matches and guess how many matches the opponent has taken. If one of the players guesses correctly, then the loser has to pay him the sum of the number held by both players. Otherwise, the pay out is zero. Write down the payoff matrix and obtain the optimal strategies of both players.
4. Explain the principle of dominance in game theory and solve the following game:

Player *A*	Player *B* B_1	B_2	B_3
A_1	1	7	2
A_2	6	2	7
A_3	5	2	6

5.

Player *A*	Player *B* B_1	B_2	B_3	B_4
A_1	8	10	9	14
A_2	10	11	8	12
A_3	13	12	14	13

6.

Player *A*	Player *B* B_1	B_2	B_3	B_4	B_5
A_1	2	4	3	8	4
A_2	5	6	3	7	8
A_3	6	7	9	8	7
A_4	4	2	8	4	3

7. In an election campaign, the strategies adopted by the ruling and opposition parties, along with payoffs, (ruling party's per cent share in votes polled) are given below:

Ruling Party's Strategies	Opposition Party's Strategies: *Campaign one days in each city*	*Campaign two day in large towns*	*Spend two day in large rural sectors*
Campaign one day in each city	55	40	35
Campaign two days in large towns	70	70	55
Spend two days in large rural sectors	75	55	65

Assume a zero-sum game. Find the optimum strategies for both parties and expected payoff to ruling party.

8. Player *A* is paid Rs 8 if two coins turn heads at the same time and Rs 10 if two coins turn tails at the same time. Player *B* is paid Rs 3 when the two coins do not match. Given the choice of being *A* or *B,* which one would you choose and what would be your strategy?
9. Even though there are several manufacturers of scooters, two firms with brand names Janata and Praja, control their market in Western India. If both manufacturers make changes in the model in the same year, their respective market shares would remain constant. Likewise, if neither makes model changes, even then their market shares would remain constant. The payoff matrix, in terms of increased/decreased percentage market share, under different possible conditions is given below:

Janata	Praja: *No change*	*Minor change*	*Major change*
No change	0	– 4	–10
Minor change	3	0	5
Major change	8	1	0

(a) Find the value of the game.
(b) What change should Janata consider if this information is available only to itself? [*Delhi Univ., MCom, 1988*]

10. A steel company is negotiating with its union for revising wages of its employees. The management, with the help of a mediator, has prepared a payoff matrix shown below. Plus sign, represents wage increase, while negative sign stands for wage decrease. The union has also constructed a table which is comparable to that developed by the management. The management does not have the specific knowledge of game theory to select the best strategy or strategies for the firm. You have to assist the management on the problem. What game value and strategies are available to the opposing group?

Additional Costs to Steel Company (Rs)

Company Strategies	Union Strategies U_1	U_2	U_3	U_4
C_1	2.50	2.70	3.50	0.20
C_2	2.00	1.60	0.80	0.80
C_3	1.40	1.20	1.50	1.30
C_4	3.00	1.40	1.90	0

11. In zero-sum two-person children's game of stone, paper and scissors, both players simultaneously call out stone, paper or scissors. The combination of paper and stone is a win of one unit for player calling paper (paper covers stone); stone and scissors is a win for stone (stone breaks scissors), and scissors and paper is a win for scissors (scissors cut paper). A call of the same item represents no payoff. Write the payoff matrix and the equivalent linear programming problem to the above game. Find the optimal strategy for both the players and the value of the game.
12. Two candidates, X and Y, are competing for the councillor's seat in a city municipal corporation. X is attempting to increase his total votes at the expense of Y. The strategies available to each candidate involve personal contacts, newspaper insertions, speeches or television appearance, and advertising. The increase

in votes available to X, given various combinations of strategies, are given below. (Assume that this is a zero-sum game, i.e. any gain of X is equal to the votes lost by Y). Determine the optimal strategies that should be adopted by X during his election campaign. How many votes should X gain by adopting the optimal strategy?

	Candidate *Y*		
Candidate *X*	*Personal contacts*	*Newspapers*	*Television*
Personal contacts	30,000	20,000	10,000
Newspapers	60,000	50,000	25,000
Television	20,000	40,000	30,000

13. Vishal who has an amount of Rs 1 lakh, is planning to invest it among three companies. He wants to buy equity shares in company A, B and C. The payoffs in terms of (i) growth in capital, and (ii) returns on the capital, are known for each of the investments under each of the three economic conditions which may prevail – recession, growth and stability. Assuming that Vishal must make his choice among the three portfolios for a period of one year in advance, his expectations of the net earning (in '000 Rs) of his Rs 1 lakh portfolio after one year is represented by the following matrix:

	Economic Condition		
Company	*Recession*	*Stability*	*Growth*
A	–15	6	10
B	4	7.5	8
C	6.5	6	15

Determine the optimal strategies for investment and the expected present return for the investor under such a policy.

[*Delhi Univ., MBA, 1996*]

14. Two firms F_1 and F_2 make colour and black & white television sets. F_1 can make either 300 colour sets in a month or an equal number of black & white sets. He makes a profit of Rs 200 on a colour set and Rs 150 on a black & white set. F_2 can, on the other hand, make either 600 colour sets or 300 colour and 300 black & white sets or 600 black & white sets per month. It also has the same profit margin on the two sets as F_1. Each month there is a market of 300 colour sets and 600 black & white sets and the manufacturers would share the market depending upon the proportion in which they manufacture a particular type of sets.

Write the payoff matrix of F_1 and F_2 per month. Obtain F_1 and F_2's optimal strategies and the value of the game.

[*Delhi Univ., MBA, 1997*]

15. Obtain the strategies for both players and the value of the game for two-person zero-sum game whose payoff matrix is given as follows:

(a)

	Player *B*		
Player *A*	B_1	B_2	B_3
A_1	1	3	11
A_2	8	5	2

(b)

	Player *B*					
Player *A*	B_1	B_2	B_3	B_4	B_5	B_6
A_1	1	3	– 1	4	2	– 5
A_2	– 3	5	6	1	2	0

16. Obtain the optimal strategies for both players and the value of the game for two-person zero-sum game whose payoff matrix is given as follows:

	Player *B*	
Player *A*	B_1	B_2
A_1	– 6	7
A_2	4	– 5
A_3	– 1	– 2
A_4	– 2	5
A_5	7	– 6

17. For the following payoff tables, transform the zero-sum game into an equivalent linear programming problem and solve it using the simplex method.

(a)

	Player *B*		
Player *A*	B_1	B_2	B_3
A_1	9	1	4
A_2	0	6	3
A_3	5	2	8

(b)

	Company *A*		
Company *B*	A_1	A_2	A_3
B_1	2	– 2	3
B_2	– 3	5	– 1

18. A soft drink company calculated the market share of two of its products against its major competitor, which has three products. The company found out the impact of additional advertisement in any one of its products against the other.

	Company *B*		
Company *A*	B_1	B_2	B_3
A_1	6	7	15
A_2	20	12	10

What is the best strategy for the company as well as the competitor? What is the payoff obtained by the company and the competitor in the long run? Use the graphical method to obtain the solution.

19. Two Firms *A* and *B* make colour and black & white television sets. Firm *A* can make either 150 colour sets in a week or an equal number of black and white sets, and make a profit of Rs 400 per colour set and Rs 300 per black & white set. Firm *B* can, on the other hand, make either 300 colour sets, or 150 colour and 150 black & white sets, or 300 black & white sets per week. It also has the same profit margin on the two sets as *A*. Each week there is a market of 150 colour sets and 300 black & white sets. The manufacturers would share market depending upon the proportion in which they manufacture a particular type of set.

Write the payoff matrix of *A* per week. Obtain, graphically, *A*'s and *B*'s optimal strategies and the value of the game.

20. In a town there are only two discount stores ABC and XYZ. Both stores run annual pre-Diwali sales. Sales are advertised through local newspapers with the aid of an advertising firm. ABC stores constructed following payoff in units of Rs 1,00,000. Find the optimal strategies for both stores and the value of the game:

	Store *XYZ*		
Store *ABC*	B_1	B_2	B_3
A_1	1	– 2	1
A_2	– 1	3	2
A_3	– 1	– 2	3

21. Assume that the two firms are competing for market share for a particular product. Each firm is considering what promotional

strategy to employ for the coming period. Assume that the following payoff matrix describes the increase in market share for Firm *A* and the decrease in market share for Firm *B*. Determine the optimal strategies for each firm.

	Firm *B*		
Firm *A*	*No promotion*	*Moderate promotion*	*Much promotion*
No promotion	5	0	–10
Moderate promotion	10	6	2
Much promotion	20	15	10

(a) Formulate a suitable linear programming model of the game, with respect to minimizing player *B*'s losses and derive the optimal strategy for *B*.

(b) Write down the dual of the above linear programming model and derive the solution to the dual from the optimal table of the primal and interpret the meaning of this solution.

22. Firm *X* is fighting for its life against the determination of firm *Y* to drive it out of the industry. Firm *X* has the choice of increasing the price, leaving it unchanged, or lowering it. Firm *Y* has the same three options. Firm *X*'s gross sales in the event of each of the pairs of choices are shown below:

	Firm *Y*'s Pricing Strategies		
Firm *X*'s Pricing Strategies	*Increase price*	*Do not change*	*Reduce price*
Increase price	90	80	110
Do not change	110	100	90
Reduce price	120	70	80

Assuming firm X as the maximizing one, formulate and solve the problem as a linear programming problem.

[*Osmania MBA, 2002*]

HINTS AND ANSWERS

1. Matching player = Non-matching player = (4/15, 11/15) and $V = -1/15$.

2. (a) Third column is dominated by the first column; in the reduced matrix third row dominates an average of first and second row.
Reduced matrix has the saddle point. Optimal strategy for player *A* is A_3 and for player is B_2 or B_3. Value of game = – 2.

(b) Player *A*: (0, 0, 2/3, 1/3); Player *B:* (0, 0, 2/3, 1/3) and $V = 8/3$.

3. The payoff matrix is:

	Player *B*	
Player *A*	2	0
	0	4

Player *A:* (2/3, 1/3); Player *B:* (2/3, l/3) and $V = 4/3$.

4. Row 3 is dominated by row 2; Column 3 in the reduced matrix is dominated by column 1.
Player *A:* (2/5, 3/5, 0), Player *B:* (1/2, 1/2, 0) and $V = 4$.

5. Column 4 is dominated by column 1. In reduced matrix, row 1 is dominated by row 3.
In reduced matrix, average of columns 2 and 3 dominate column 1. The reduced matrix has a saddle point. Optimal strategy for player *A* is A_3 (row 3) and for player *B* is B_1 (column 1). The value of game is 12.

6. Optimal strategy for player *A* is A_3 and for player *B* is B_1. The value of game is 6.

7.

	B_1	B_2	B_3
A_1	55	40	30
A_2	70	70	55
A_3	75	55	65

For ruling party: (0, 2/5, 3/5)
For opposition party: (0, 2/5, 3/5)
Expected value of game = 61

8.

	Player *B*		
Player *A*	$B_1(H)$	$B_2(T)$	
A_1 (H)	8	– 3	For player A : (4/15, 11/15)
A_2 (T)	– 3	1	For player B : (4/15, 11/15)
			Expected value of game = 1/15

9. (a)

	Praja	
Janata	B_1	B_2
A_2	0	5
A_3	1	0

For Janata: (0, 1/6, 5/6)
For Praja: (0, 5/6, 1/6)
Expected value of game = 5/6

(b) Janata may consider to have minor change with probability 5/6 and of major change with probability 1/6.

10. First interchange rows into columns to recast payoff matrix because 'Union' is considered as 'maximizing' player.
The game has no saddle point. Delete row 4 (dominated by row 3) and then delete column 1 (dominated by columns 2 and 3) and column 4 (dominated by column 3). The reduced payoff matrix is

	C_2	C_3	
U_1	2.00	1.40	1.50 – 0.80 = 0.70, $p(U_1) = 7/13$
U_3	0.80	1.50	2.00 – 1.40 = 0.6, $p(U_3) = 6/13$

1.50 – 1.40 = 0.10 2.00 – 0.80 = 1.20

$p(C_2) = 1/13$, $p(C_3) = 12/13$; Value of the game = 18.8/13.

15. (a) Player *A*: (3/11, 8/11), Player *B*: (0, 2/11, 9/11) and $V = 49/11$

(b) Player *A*: (4/5, 1/5), Player *B*: (0, 3/5, 0, 2/5, 0, 0,) and $V = 17/5$

16. Player *A*: (0, 0, 0, 13/20, 7/20), Player *B*: (11/20, 9/20) and $V = 23/20$.

17. (a) Player *A*: (3/8, 13/24, 1/12), Player *B*: (7/24, 5/9, 11/72) and $V = 91/24$

(b) *A* constant, $C = 4$ is added to all the elements of the payoff matrix.

18. Company *A*: (2/3, 1/3, 0), Company *B*: (7/12, 5/12) and $V = 1/3$

19.

Firm A	Firm B		
	B_1	B_2	B_3
A_1	20,000	30,000	60,000
A_2	45,000	45,000	30,000

A_1 : 150 colour sets;
A_2 : 150 black & white sets
B_1 : 300 colour sets;
B_2 : 150 colour and 150 black & white sets;
B_3 : 300 black & white sets.

Profit to Firm A for each combination of its strategies with strategies of Firm B is calculated as:

For combination A_1B_1 : Market share × Total market × Profit per set

$$: \frac{50}{150+300} \times 150 \times 400 = 20{,}000$$

Firm A: (3/11, 8/11); Firm B: (6/11, 0, 5/11),

V = Rs 38,182 approx.

20. Add 2 (absolute value of the smallest negative payoff value) to each element of the payoff matrix. Then formulate an LP model for store XYZ.

Optimal values of decision variables are: $y_1 = y_2 = y_3 = 1/6$ and $Z = 1/2 = 1/V$ or $V = 2$. Subtract 2 from $V = 2$ to get $V = 0$.

CHAPTER SUMMARY

The theory of games provides mathematical models that are useful in explaining interactive decision-making process, where two or more competitions (players) are involved under condition of conflict and competition.

A two-person game allows only two competitors to be involved in the game. In such a game if the sum of gains of one player is equal to the sum of losses of another, then the game is said to be *two-person zero-sum* game.

For a payoff matrix without saddle point, a number of solution methods can be used to find the value of the game and probabilities of using strategies by both players.

CHAPTER CONCEPTS QUIZ

True or False

1. In a two person zero-sum game, a saddle point always exists.
2. If there are only two strategies, the payoff matrix has a saddle point.
3. In a pure strategy game, each player always plays just one strategy.
4. In a mixed strategy game, each player random by chooses the strategy to be used.
5. A mixed strategy game is based on the assumption that players act irrationally.
6. The strategy for a players is a course of action that the adopts for each payoff.
7. The value of the game is the expected outcome per play when players follow their optimal strategy.
8. The payoff is a quantitative measure of satisfaction that a player gets at the end of the play.
9. If the maximin value is same as the minimax value, than the game is said to have a saddle point.
10. A game said to be strictly determinable if lower and upper values of the game are equal.

Fill in the Blanks

11. A game is said be _____ if lower and upper values of the game are same as well as zero.
12. A course of action that puts any player in the most preferred position irrespective of the course of action chosen by the competitor is called _____.
13. If a game involves more than two players, then it is called a _____ game.
14. The pure strategy on the only course of action that is _____ being chosen by a player.
15. The rules of _____ are used to reduce the size of the payoff matrix.
16. The _____ is used to determine the probability of using different strategies by both the players.
17. The _____ is used to find optimal strategy for each player in a payoff matrix of size 2 × 2 without saddle point.
18. The _____ is useful for game where payoff matrix is of size 2 × n or m × 2.
19. The _____ method is useful for game with larger payoff matrix without saddle point.
20. The concept of _____ is used in simplifying the solution of a rectangular game.

Multiple Choice

21. Two person zero-sum game means that the
    (a) sum of losses to one player is equal to the sum of gains to other
    (b) sum of losses to one player is not equal to the sum of gains to other
    (c) both (a) and (b)
    (d) none of the above
22. Game theory models are classified by the
    (a) number of players
    (b) sum of all payoffs
    (c) number of strategies
    (d) all of the above
23. A game is said to be fair if
    (a) both upper and lower values of the game are same and zero
    (b) upper and lower values of the game are not equal
    (c) upper value is more than lower value of the game
    (d) none of the above

24. What happens when maximin and minimax values of the game are same?
    (a) no solution exists
    (b) solution is mixed
    (c) saddle point exists
    (d) none of the above
25. A mixed strategy game can be solved by
    (a) algebraic method
    (b) matrix method
    (c) graphical method
    (d) all of the above
26. The size of the payoff matrix of a game can be reduced by using the principle of
    (a) game inversion
    (b) rotation reduction
    (c) dominance
    (d) game transpose
27. The payoff value for which each player in a game always selects the same strategy is called the
    (a) saddle point
    (b) equilibrium point
    (c) both (a) and (b)
    (d) none of the above
28. Games which involve more than two players are called
    (a) conflicting games
    (b) negotiable games
    (c) N-person games
    (d) all of the above
29. When the sum of gains of one player is equal to the sum of losses to another player in a game, this situation is known as
    (a) biased game
    (b) zero-sum game
    (c) fair game
    (d) all of the above
30. When no saddle point is found in a payoff matrix of a game, the value of the game is then found by
    (a) knowing joint probabilities of each row and column combination to calculate expected payoff for that combination and adding all such values
    (b) reducing size of the game to apply algebraic method
    (c) both (a) and (b)
    (d) none of the above
31. A saddle point exists when
    (a) maximin value = maximax value
    (b) minimax value = minimum value
    (c) minimax value = maximin value
    (d) none of the above
32. In a pure strategy game
    (a) any strategy may be selected arbitrarily
    (b) a particular strategy is selected by each player
    (c) both players select their optimal strategy
    (d) none of the above
33. In a mixed strategy game
    (a) no saddle point exist
    (b) each player selects the same strategy without considering other player's choice
    (c) each player always selects same strategy
    (d) all of the above
34. Linear programming method should be used to determine value of the game when size of payoff matrix is
    (a) 2×2 (b) 3×4
    (c) $m \times 2$ (d) $2 \times n$
35. Game theory is the study of
    (a) selecting optimal strategies
    (b) resolving conflict between players
    (c) both (a) and (b)
    (d) none of the above

Answers to Quiz

1. F	2. F	3. T	4. T	5. F	6. T	7. T	8. T	9. T	10. T
11. fair game		12. optimal strategy		13. unperson		14. ****		15. dominance	
16. algebraic method		17. short-cut method		18. graphical method		19. LP	20. dominance		
21. (a)	22. (d)	23. (a)	24. (c)	25. (d)	26. (c)	27. (c)	28. (c)	29. (b)	30. (c)
31. (c)	32. (c)	33. (a)	34. (b)	35. (a)					